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11. Long-Range Out-of-Sample Properties of Autoregressive Neural Networks

Patrick L. Leoni

https://doi.org/10.1162/neco.2009.09-07-615 | pp. 1-8

We consider already-trained discrete autoregressive neural networks in their most general representations, with the exclusion of time-varying input though, and we provide tight sufficient conditions and elementary proofs for the existence of an attractor, uniqueness, and global convergence. Those conditions can be used as easy-to-check criteria when convergence (or not) of long-range predictions is desirable.

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12. Spike-Timing Error Backpropagation in Theta Neuron Networks

Sam McKennoch, Thomas Voegtlin, Linda Bushnell

https://doi.org/10.1162/neco.2009.09-07-610 | pp. 9-45

The main contribution of this letter is the derivation of a steepest gradient descent learning rule for a multilayer network of theta neurons, a one-dimensional nonlinear neuron model. Central to our model is the

assumption that the intrinsic neuron dynamics are sufficient to achieve consistent time coding, with no need to involve the precise shape of postsynaptic currents; this assumption departs from other related models such as SpikeProp and Tempotron learning. Our results clearly show that it is possible to perform complex computations by applying supervised learning techniques to the spike times and time response properties of nonlinear integrate and fire neurons. Networks trained with our multilayer training rule are shown to have similar generalization abilities for spike latency pattern classification as Tempotron learning. The rule is also able to train networks to perform complex regression tasks that neither SpikeProp or Tempotron learning appears to be capable of.

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13. A Master Equation Formalism for Macroscopic Modeling of Asynchronous Irregular Activity States

Sami El■Boustani, Alain Destexhe

https://doi.org/10.1162/neco.2009.02-08-710 | pp. 46-100

Many efforts have been devoted to modeling asynchronous irregular (AI) activity states, which resemble the complex activity states seen in the cerebral cortex of awake animals. Most of models have considered balanced networks of excitatory and inhibitory spiking neurons in which AI states are sustained through recurrent sparse connectivity, with or without external input. In this letter we propose a mesoscopic description of such AI states. Using master equation formalism, we derive a second-order mean-field set of ordinary differential equations describing the temporal evolution of randomly connected balanced networks. This formalism takes into account finite size effects and is applicable to any neuron model as long as its transfer function can be characterized. We compare the predictions of this approach with numerical simulations for different network configurations and parameter spaces. Considering the randomly connected network as a unit, this approach could be used to build large-scale networks of such connected units, with an aim to model activity states constrained by macroscopic measurements, such as voltage-sensitive dye imaging.

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14. A New Approach for Estimating the Attraction Domain for Hopfield-Type Neural Networks

Dequan Jin, Jigen Peng

https://doi.org/10.1162/neco.2009.11-07-637 | pp. 101-120

In this letter, using methods proposed by E. Kaslik, St. Balint, and their colleagues, we develop a new method, expansion approach, for estimating the attraction domain of asymptotically stable equilibrium points of Hopfield-type neural networks. We prove theoretically and demonstrate numerically that the proposed approach is feasible and efficient. The numerical results that obtained in the application examples, including the network system considered by E. Kaslik, L. Br■escu, and St. Balint, indicate that the proposed approach is able to achieve better attraction domain estimation.

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15. Density-Weighted Nyström Method for Computing Large Kernel Eigensystems

Kai Zhang, James T. Kwok
https://doi.org/10.1162/neco.2009.11-07-651 | pp. 121-146

The Nyström method is a well-known sampling-based technique for approximating the eigensystem of large kernel matrices. However, the chosen samples in the Nyström method are all assumed to be of equal importance, which deviates from the integral equation that defines the kernel eigenfunctions. Motivated by this observation, we extend the Nyström method to a more general, density-weighted version. We show that by introducing the probability density function as a natural weighting scheme, the approximation of the eigensystem can be greatly improved. An efficient algorithm is proposed to enforce such weighting in practice, which has the same complexity as the original Nyström method and hence is notably cheaper than several other alternatives. Experiments on kernel principal component analysis, spectral clustering, and image segmentation demonstrate the encouraging performance of our algorithm.

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16. Persistent Neural States: Stationary Localized Activity Patterns in Nonlinear Continuous n-Population, q-Dimensional Neural Networks

Olivier Faugeras, Romain Veltz, François Grimbert
https://doi.org/10.1162/neco.2009.12-07-660 | pp. 147-187

Neural continuum networks are an important aspect of the modeling of macroscopic parts of the cortex. Two classes of such networks are considered: voltage and activity based. In both cases, our networks contain an arbitrary number, n, of interacting neuron populations. Spatial nonsymmetric connectivity functions represent cortico-cortical, local connections, and external inputs represent nonlocal connections. Sigmoidal nonlinearities model the relationship between (average) membrane potential and activity. Departing from most of the previous work in this area, we do not assume the nonlinearity to be singular, that is, represented by the discontinuous Heaviside function. Another important difference from previous work is that we relax the assumption that the domain of definition where we study these networks is infinite, that is, equal to [Formula: see text] or [Formula: see text]. We explicitly consider the biologically more relevant case of a bounded subset Ω of [Formula: see text], a better model of a piece of cortex. The time behavior of these networks is described by systems of integro-differential equations. Using methods of functional analysis, we study the existence and uniqueness of a stationary (i.e., time-independent) solution of these equations in the case of a stationary input. These solutions can be seen as ‘persistent’; they are also sometimes called bumps. We show that under very mild assumptions on the connectivity functions and because we do not use the Heaviside function for the nonlinearities, such solutions always exist. We also give sufficient conditions on the connectivity functions for the solution to be absolutely stable, that is, independent of the initial state of the network. We then study the sensitivity of the solutions to variations of such parameters as the connectivity functions, the sigmoids, the external inputs, and, last but not least, the shape of the domain of existence Ω of the neural continuum networks. These theoretical results are illustrated and corroborated by a large number of numerical experiments in most of the cases 2 ■ n ■ 3, 2 ■ q ■ 3.

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17. Generation of Correlated Spike Trains

Romain Brette
https://doi.org/10.1162/neco.2009.12-07-657 | pp. 188-215

Neuronal spike trains display correlations at diverse timescales throughout the nervous system. The functional significance of these correlations is largely unknown, and computational investigations can help us understand their role. In order to generate correlated spike trains with given statistics, several case-specific methods have been described in the litterature. This letter presents two general methods to generate sets of spike trains with given firing rates and pairwise correlation functions, along with efficient simulation algorithms.

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18. Getting to Know Your Neighbors: Unsupervised Learning of Topography from Real-World, Event-Based Input

Martin Boerlin, Tobi Delbrück, Kynan Eng

https://doi.org/10.1162/neco.2009.06-07-554 | pp. 216-238

Biological neural systems must grow their own connections and maintain topological relations between elements that are related to the sensory input surface. Artificial systems have traditionally prewired such maps, but the sensor arrangement is not always known and can be expensive to specify before run time. Here we present a method for learning and updating topographic maps in systems comprising modular, event-based elements. Using an unsupervised neural spike-timing-based learning rule combined with Hebbian learning, our algorithm uses the spatiotemporal coherence of the external world to train its network. It improves on existing algorithms by not assuming a known topography of the target map and includes a novel method for automatically detecting edge elements. We show how, for stimuli that are small relative to the sensor resolution, the temporal learning window parameters can be determined without using any user-specified constants. For stimuli that are larger relative to the sensor resolution, we provide a parameter extraction method that generally outperforms the small-stimulus method but requires one user-specified constant. The algorithm was tested on real data from a 64 x 64-pixel section of an event-based temporal contrast silicon retina and a 360-tile tactile luminous floor. It learned 95.8% of the correct neighborhood relations for the silicon retina within about 400 seconds of real-world input from a driving scene and 98.1% correct for the sensory floor after about 160 minutes of human pedestrian traffic. Residual errors occurred in regions receiving little or ambiguous input, and the learned topological representations were able to update automatically in response to simulated damage. Our algorithm has applications in the design of modular autonomous systems in which the interfaces between components are learned during operation rather than at design time.

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19. Decision-Theoretic Saliency: Computational Principles, Biological Plausibility, and Implications for Neurophysiology and Psychophysics

Dashan Gao, Nuno Vasconcelos

https://doi.org/10.1162/neco.2009.11-06-391 | pp. 239-271

A decision-theoretic formulation of visual saliency, first proposed for top-down processing (object recognition) (Gao & Vasconcelos, 2005a), is extended to the problem of bottom-up saliency. Under this formulation, optimality is defined in the minimum probability of error sense, under a constraint of computational parsimony. The saliency of the visual features at a given location of the visual field is defined as the power of those features to discriminate between the stimulus at the location and a null hypothesis. For bottom-up saliency, this is the set of visual features that surround the location under consideration.

Discrimination is defined in an information-theoretic sense and the optimal saliency detector derived for a class of stimuli that complies with known statistical properties of natural images. It is shown that under the assumption that saliency is driven by linear filtering, the optimal detector consists of what is usually referred to as the standard architecture of V1: a cascade of linear filtering, divisive normalization, rectification, and spatial pooling. The optimal detector is also shown to replicate the fundamental properties of the psychophysics of saliency: stimulus pop-out, saliency asymmetries for stimulus presence versus absence, disregard of feature conjunctions, and Weber's law. Finally, it is shown that the optimal saliency architecture can be applied to the solution of generic inference problems. In particular, for the class of stimuli studied, it performs the three fundamental operations of statistical inference: assessment of probabilities, implementation of Bayes decision rule, and feature selection.

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20. Prototype Classification: Insights from Machine Learning

Arnulf Graf, Olivier Bousquet, Gunnar Rätsch, Bernhard Schölkopf

https://doi.org/10.1162/neco.2009.01-07-443 | pp. 272-300

We shed light on the discrimination between patterns belonging to two different classes by casting this decoding problem into a generalized prototype framework. The discrimination process is then separated into two stages: a projection stage that reduces the dimensionality of the data by projecting it on a line and a threshold stage where the distributions of the projected patterns of both classes are separated. For this, we extend the popular mean-of-class prototype classification using algorithms from machine learning that satisfy a set of invariance properties. We report a simple yet general approach to express different types of linear classification algorithms in an identical and easy-to-visualize formal framework using generalized prototypes where these prototypes are used to express the normal vector and offset of the hyperplane. We investigate non-margin classifiers such as the classical prototype classifier, the Fisher classifier, and the relevance vector machine. We then study hard and soft margin classifiers such as the support vector machine and a boosted version of the prototype classifier. Subsequently, we relate mean-of-class prototype classification to other classification algorithms by showing that the prototype classifier is a limit of any soft margin classifier and that boosting a prototype classifier yields the support vector machine. While giving novel insights into classification per se by presenting a common and unified formalism, our generalized prototype framework also provides an efficient visualization and a principled comparison of machine learning classification.

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21. A Spiking Neural Network Model of an Actor-Critic Learning Agent

Wiebke Potjans, Abigail Morrison, Markus Diesmann

https://doi.org/10.1162/neco.2008.08-07-593 | pp. 301-339

The ability to adapt behavior to maximize reward as a result of interactions with the environment is crucial for the survival of any higher organism. In the framework of reinforcement learning, temporal-difference learning algorithms provide an effective strategy for such goal-directed adaptation, but it is unclear to what extent these algorithms are compatible with neural computation. In this article, we present a spiking neural network model that implements actor-critic temporal-difference learning by combining local plasticity rules with a global reward signal. The network is capable of solving a nontrivial gridworld task with sparse rewards. We derive a quantitative mapping of plasticity parameters and synaptic weights to the

corresponding variables in the standard algorithmic formulation and demonstrate that the network learns with a similar speed to its discrete time counterpart and attains the same equilibrium performance.

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22. A Gradient Learning Rule for the Tempotron

Robert Urbanczik, Walter Senn

https://doi.org/10.1162/neco.2008.09-07-605 | pp. 340-352

We introduce a new supervised learning rule for the tempotron task: the binary classification of input spike trains by an integrate-and-fire neuron that encodes its decision by firing or not firing. The rule is based on the gradient of a cost function, is found to have enhanced performance, and does not rely on a specific reset mechanism in the integrate-and-fire neuron.

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23. Simplicity and Efficiency of Integrate-and-Fire Neuron Models

Hans Ekkehard Pleßer, Markus Diesmann

https://doi.org/10.1162/neco.2008.03-08-731 | pp. 353-359

Lovelace and Cios (2008) recently proposed a very simple spiking neuron (VSSN) model for simulations of large neuronal networks as an efficient replacement for the integrate-and-fire neuron model. We argue that the VSSN model falls behind key advances in neuronal network modeling over the past 20 years, in particular, techniques that permit simulators to compute the state of the neuron without repeated summation over the history of input spikes and to integrate the subthreshold dynamics exactly. State-of-the-art solvers for networks of integrate-and-fire model neurons are substantially more efficient than the VSSN simulator and allow routine simulations of networks of some 10(5) neurons and 10(9) connections on moderate computer clusters.

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24. Spike Train Statistics and Dynamics with Synaptic Input from any Renewal Process: A Population Density Approach

Cheng Ly, Daniel Tranchina

https://doi.org/10.1162/neco.2008.03-08-743 | pp. 360-396

In the probability density function (PDF) approach to neural network modeling, a common simplifying assumption is that the arrival times of elementary postsynaptic events are governed by a Poisson process. This assumption ignores temporal correlations in the input that sometimes have important physiological consequences. We extend PDF methods to models with synaptic event times governed by any modulated renewal process. We focus on the integrate-and-fire neuron with instantaneous synaptic kinetics and a random elementary excitatory postsynaptic potential (EPSP), A. Between presynaptic events, the membrane voltage, v, decays exponentially toward rest, while s, the time since the last synaptic input event, evolves with unit velocity. When a synaptic event arrives, v jumps by A, and s is reset to zero. If v crosses the threshold voltage, an action potential occurs, and v is reset to v(reset). The probability per unit time of a synaptic event at time t, given the elapsed time s since the last event, h(s, t), depends on specifics of the renewal process. We study how regularity of the train of synaptic input events affects output spike rate, PDF and coefficient of variation (CV) of the interspike interval, and the autocorrelation function of the output spike

train. In the limit of a deterministic, clocklike train of input events, the PDF of the interspike interval converges to a sum of delta functions, with coefficients determined by the PDF for A. The limiting autocorrelation function of the output spike train is a sum of delta functions whose coefficients fall under a damped oscillatory envelope. When the EPSP CV, sigma A/mu A, is equal to 0.45, a CV for the intersynaptic event interval, sigma T/mu T = 0.35, is functionally equivalent to a deterministic periodic train of synaptic input events (CV = 0) with respect to spike statistics. We discuss the relevance to neural network simulations.

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25. Generating Spike Trains with Specified Correlation Coefficients

Jakob H. Macke, Philipp Berens, Alexander S. Ecker, Andreas S. Tolias, Matthias Bethge
https://doi.org/10.1162/neco.2008.02-08-713 | pp. 397-423

Spike trains recorded from populations of neurons can exhibit substantial pairwise correlations between neurons and rich temporal structure. Thus, for the realistic simulation and analysis of neural systems, it is essential to have efficient methods for generating artificial spike trains with specified correlation structure. Here we show how correlated binary spike trains can be simulated by means of a latent multivariate gaussian model. Sampling from the model is computationally very efficient and, in particular, feasible even for large populations of neurons. The entropy of the model is close to the theoretical maximum for a wide range of parameters. In addition, this framework naturally extends to correlations over time and offers an elegant way to model correlated neural spike counts with arbitrary marginal distributions.

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26. A Reproducing Kernel Hilbert Space Framework for Spike Train Signal Processing

António R. C. Paiva, Il Memming Park, José C. Pr■■ncipe
https://doi.org/10.1162/neco.2008.09-07-614 | pp. 424-449

This letter presents a general framework based on reproducing kernel Hilbert spaces (RKHS) to mathematically describe and manipulate spike trains. The main idea is the definition of inner products to allow spike train signal processing from basic principles while incorporating their statistical description as point processes. Moreover, because many inner products can be formulated, a particular definition can be crafted to best fit an application. These ideas are illustrated by the definition of a number of spike train inner products. To further elicit the advantages of the RKHS framework, a family of these inner products, the cross-intensity (CI) kernels, is analyzed in detail. This inner product family encapsulates the statistical description from the conditional intensity functions of spike trains. The problem of their estimation is also addressed. The simplest of the spike train kernels in this family provide an interesting perspective to others' work, as will be demonstrated in terms of spike train distance measures. Finally, as an application example, the RKHS framework is used to derive a clustering algorithm for spike trains from simple principles.

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27. Identifying Functional Connectivity in Large-Scale Neural Ensemble Recordings: A Multiscale Data Mining Approach

Seif Eldawlatly, Rong Jin, Karim Oweiss

https://doi.org/10.1162/neco.2008.09-07-606 | pp. 450-477

Identifying functional connectivity between neuronal elements is an essential first step toward understanding how the brain orchestrates information processing at the single-cell and population levels to carry out biological computations. This letter suggests a new approach to identify functional connectivity between neuronal elements from their simultaneously recorded spike trains. In particular, we identify clusters of neurons that exhibit functional interdependency over variable spatial and temporal patterns of interaction. We represent neurons as objects in a graph and connect them using arbitrarily defined similarity measures calculated across multiple timescales. We then use a probabilistic spectral clustering algorithm to cluster the neurons in the graph by solving a minimum graph cut optimization problem. Using point process theory to model population activity, we demonstrate the robustness of the approach in tracking a broad spectrum of neuronal interaction, from synchrony to rate co-modulation, by systematically varying the length of the firing history interval and the strength of the connecting synapses that govern the discharge pattern of each neuron. We also demonstrate how activity-dependent plasticity can be tracked and quantified in multiple network topologies built to mimic distinct behavioral contexts. We compare the performance to classical approaches to illustrate the substantial gain in performance.

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28. State-Dependent Computation Using Coupled Recurrent Networks

Ueli Rutishauser, Rodney J. Douglas

https://doi.org/10.1162/neco.2008.03-08-734 | pp. 478-509

Although conditional branching between possible behavioral states is a hallmark of intelligent behavior, very little is known about the neuronal mechanisms that support this processing. In a step toward solving this problem, we demonstrate by theoretical analysis and simulation how networks of richly interconnected neurons, such as those observed in the superficial layers of the neocortex, can embed reliable, robust finite state machines. We show how a multistable neuronal network containing a number of states can be created very simply by coupling two recurrent networks whose synaptic weights have been configured for soft winner-take-all (sWTA) performance. These two sWTAs have simple, homogeneous, locally recurrent connectivity except for a small fraction of recurrent cross-connections between them, which are used to embed the required states. This coupling between the maps allows the network to continue to express the current state even after the input that elicited that state is withdrawn. In addition, a small number of transition neurons implement the necessary input-driven transitions between the embedded states. We provide simple rules to systematically design and construct neuronal state machines of this kind. The significance of our finding is that it offers a method whereby the cortex could construct networks supporting a broad range of sophisticated processing by applying only small specializations to the same generic neuronal circuit.

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29. The Problem of Rapid Variable Creation

Robert F. Hadley

https://doi.org/10.1162/neco.2008.07-07-572 | pp. 510-532

Both Marcus (2001) and Jackendoff (2002) have emphasized the importance of finding credible explanations for the occurrence of variables within cognitive representations. Marcus, in particular, has argued that a prevailing form of connectionist modeling, eliminative connectionism, cannot adequately explain crucial forms of human generalization. Eliminative connectionism eschews the use of explicitly

represented variables, and the latter, Marcus contends, play an essential role in the forms of generalization that he considers. Recently, van der Velde and de Kamps (2006) proposed a neural blackboard architecture, which they assert to have satisfied the variable representation needs that Marcus and Jackendoff identified. However, this letter argues that closely related variants of Marcus's generalization examples possess variable requirements that are incompatible with the van der Velde and de Kamps approach. Moreover, it is argued here that these newly proposed variants present a severe challenge not only for eliminative connectionism but for all network training methods that require iterative tuning of synaptic strengths. The letter focuses on generalization cases that necessitate either virtually instantaneous creation of variables or very rapid deployment of preexisting variables within highly novel contexts.

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30. Nonparametric Conditional Density Estimation Using Piecewise-Linear Solution Path of Kernel Quantile Regression

Ichiro Takeuchi, K. Nomura, Takafumi Kanamori

https://doi.org/10.1162/neco.2008.10-07-628 | pp. 533-559

The goal of regression analysis is to describe the stochastic relationship between an input vector x and a scalar output y. This can be achieved by estimating the entire conditional density p(y / x). In this letter, we present a new approach for nonparametric conditional density estimation. We develop a piecewise-linear path-following method for kernel-based quantile regression. It enables us to estimate the cumulative distribution function of p(y / x) in piecewise-linear form for all x in the input domain. Theoretical analyses and experimental results are presented to show the effectiveness of the approach.

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31. Arbitrary Norm Support Vector Machines

Kaizhu Huang, Danian Zheng, Irwin King, Michael R. Lyu

https://doi.org/10.1162/neco.2008.12-07-667 | pp. 560-582

Support vector machines (SVM) are state-of-the-art classifiers. Typically L2-norm or L1-norm is adopted as a regularization term in SVMs, while other norm-based SVMs, for example, the L0-norm SVM or even the L(infinity)-norm SVM, are rarely seen in the literature. The major reason is that L0-norm describes a discontinuous and nonconvex term, leading to a combinatorially NP-hard optimization problem. In this letter, motivated by Bayesian learning, we propose a novel framework that can implement arbitrary norm-based SVMs in polynomial time. One significant feature of this framework is that only a sequence of sequential minimal optimization problems needs to be solved, thus making it practical in many real applications. The proposed framework is important in the sense that Bayesian priors can be efficiently plugged into most learning methods without knowing the explicit form. Hence, this builds a connection between Bayesian learning and the kernel machines. We derive the theoretical framework, demonstrate how our approach works on the L0-norm SVM as a typical example, and perform a series of experiments to validate its advantages. Experimental results on nine benchmark data sets are very encouraging. The implemented L0-norm is competitive with or even better than the standard L2-norm SVM in terms of accuracy but with a reduced number of support vectors, -9.46% of the number on average. When compared with another sparse model, the relevance vector machine, our proposed algorithm also demonstrates better sparse properties with a training speed over seven times faster.

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32. Sequential Triangle Strip Generator Based on Hopfield Networks

Ji■■■ Š■■ma, Radim Ln■ni■ka

https://doi.org/10.1162/neco.2008.10-07-623 | pp. 583-617

The important task of generating the minimum number of sequential triangle strips (tristrips) for a given triangulated surface model is motivated by applications in computer graphics. This hard combinatorial optimization problem is reduced to the minimum energy problem in Hopfield nets by a linear-size construction. In particular, the classes of equivalent optimal stripifications are mapped one to one to the minimum energy states reached by a Hopfield network during sequential computation starting at the zero initial state. Thus, the underlying Hopfield network powered by simulated annealing (i.e., Boltzmann machine), which is implemented in the program HTGEN, can be used for computing the semioptimal stripifications. Practical experiments confirm that one can obtain much better results using HTGEN than by a leading conventional stripification program FTSG (a reference stripification method not based on neural nets), although the running time of simulated annealing grows rapidly near the global optimum. Nevertheless, HTGEN exhibits empirical linear time complexity when the parameters of simulated annealing (i.e., the initial temperature and the stopping criterion) are fixed and thus provides the semioptimal offline solutions, even for huge models of hundreds of thousands of triangles, within a reasonable time.

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33. Sequential Optimal Design of Neurophysiology Experiments

Jeremy Lewi, Robert J. Butera, Liam Paninski

https://doi.org/10.1162/neco.2008.08-07-594 | pp. 619-687

Adaptively optimizing experiments has the potential to significantly reduce the number of trials needed to build parametric statistical models of neural systems. However, application of adaptive methods to neurophysiology has been limited by severe computational challenges. Since most neurons are high-dimensional systems, optimizing neurophysiology experiments requires computing high-dimensional integrations and optimizations in real time. Here we present a fast algorithm for choosing the most informative stimulus by maximizing the mutual information between the data and the unknown parameters of a generalized linear model (GLM) that we want to fit to the neuron's activity. We rely on important log concavity and asymptotic normality properties of the posterior to facilitate the required computations. Our algorithm requires only low-rank matrix manipulations and a two-dimensional search to choose the optimal stimulus. The average running time of these operations scales quadratically with the dimensionality of the GLM, making real-time adaptive experimental design feasible even for high-dimensional stimulus and parameter spaces. For example, we require roughly 10 milliseconds on a desktop computer to optimize a 100-dimensional stimulus. Despite using some approximations to make the algorithm efficient, our algorithm asymptotically decreases the uncertainty about the model parameters at a rate equal to the maximum rate predicted by an asymptotic analysis. Simulation results show that picking stimuli by maximizing the mutual information can speed up convergence to the optimal values of the parameters by an order of magnitude compared to using random (nonadaptive) stimuli. Finally, applying our design procedure to real neurophysiology experiments requires addressing the nonstationarities that we would expect to see in neural responses; our algorithm can efficiently handle both fast adaptation due to spike history effects and slow, nonsystematic drifts in a neuron's activity.

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34. Information in the Nonstationary Case

Vincent Q. Vu, Bin Yu, Robert E. Kass

https://doi.org/10.1162/neco.2008.01-08-700 | pp. 688-703

Information estimates such as the direct method of Strong, Koberle, de Ruyter van Steveninck, and Bialek (1998) sidestep the difficult problem of estimating the joint distribution of response and stimulus by instead estimating the difference between the marginal and conditional entropies of the response. While this is an effective estimation strategy, it tempts the practitioner to ignore the role of the stimulus and the meaning of mutual information. We show here that as the number of trials increases indefinitely, the direct (or plug-in) estimate of marginal entropy converges (with probability 1) to the entropy of the time-averaged conditional distribution of the response, and the direct estimate of the conditional entropy converges to the time-averaged entropy of the conditional distribution of the response. Under joint stationarity and ergodicity of the response and stimulus, the difference of these quantities converges to the mutual information. When the stimulus is deterministic or nonstationary the direct estimate of information no longer estimates mutual information, which is no longer meaningful, but it remains a measure of variability of the response distribution across time.

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35. A Generalized Linear Integrate-and-Fire Neural Model Produces Diverse Spiking Behaviors

■*tefan Mihala*■*, Ernst Niebur*

https://doi.org/10.1162/neco.2008.12-07-680 | pp. 704-718

For simulations of neural networks, there is a trade-off between the size of the network that can be simulated and the complexity of the model used for individual neurons. In this study, we describe a generalization of the leaky integrate-and-fire model that produces a wide variety of spiking behaviors while still being analytically solvable between firings. For different parameter values, the model produces spiking or bursting, tonic, phasic or adapting responses, depolarizing or hyperpolarizing after potentials and so forth. The model consists of a diagonalizable set of linear differential equations describing the time evolution of membrane potential, a variable threshold, and an arbitrary number of firing-induced currents. Each of these variables is modified by an update rule when the potential reaches threshold. The variables used are intuitive and have biological significance. The model's rich behavior does not come from the differential equations, which are linear, but rather from complex update rules. This single-neuron model can be implemented using algorithms similar to the standard integrate-and-fire model. It is a natural match with event-driven algorithms for which the firing times are obtained as a solution of a polynomial equation.

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36. Complete Stability in Multistable Delayed Neural Networks

Chang■*Yuan Cheng, Chih-Wen Shih*

https://doi.org/10.1162/neco.2008.03-07-492 | pp. 719-740

We investigate the complete stability for multistable delayed neural networks. A new formulation modified from the previous studies on multistable networks is developed to derive componentwise dynamical property. An iteration argument is then constructed to conclude that every solution of the network converges to a single equilibrium as time tends to infinity. The existence of 3n equilibria and 2n positively invariant sets

for the n-neuron system remains valid under the new formulation. The theory is demonstrated by a numerical illustration.

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37. An Oscillatory Hebbian Network Model of Short-Term Memory

Ransom K. Winder, James A. Reggia, Scott A. Weems, Michael F. Bunting

https://doi.org/10.1162/neco.2008.02-08-715 | pp. 741-761

Recurrent neural architectures having oscillatory dynamics use rhythmic network activity to represent patterns stored in short-term memory. Multiple stored patterns can be retained in memory over the same neural substrate because the network's state persistently switches between them. Here we present a simple oscillatory memory that extends the dynamic threshold approach of Horn and Usher (1991) by including weight decay. The modified model is able to match behavioral data from human subjects performing a running memory span task simply by assuming appropriate weight decay rates. The results suggest that simple oscillatory memories incorporating weight decay capture at least some key properties of human short-term memory. We examine the implications of the results for theories about the relative role of interference and decay in forgetting, and hypothesize that adjustments of activity decay rate may be an important aspect of human attentional mechanisms.

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38. Computational Predictions on the Receptive Fields and Organization of V2 for Shape Processing

Yiu Fai Sit, Risto Miikkulainen

https://doi.org/10.1162/neco.2008.01-07-431 | pp. 762-785

It has been more than 40 years since the first studies of the secondary visual cortex (V2) were published. However, no concrete hypothesis on how the receptive field of V2 neurons supports general shape processing has been proposed to date. Using a computational model that follows the principle of self-organization, we advance two hypotheses in this letter: (1) typical V2 orientation-selective receptive field contains a primary orientation and a secondary orientation component, forming a corner, a junction, or a cross; and (2) V2 columns with the same primary orientation form contiguous domains, divided into subdomains that prefer different secondary orientations. The first hypothesis is consistent with existing experimental evidence, and both hypotheses can be tested with current techniques in animals. In this manner, computational modeling can be used to provide verifiable predictions that eventually allow us to understand the role of V2 in visual processing.

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39. The Variational Gaussian Approximation Revisited

Manfred Opper, Cédric Archambeau

https://doi.org/10.1162/neco.2008.08-07-592 | pp. 786-792

The variational approximation of posterior distributions by multivariate gaussians has been much less popular in the machine learning community compared to the corresponding approximation by factorizing distributions. This is for a good reason: the gaussian approximation is in general plagued by an Omicron(N)(2) number of variational parameters to be optimized, N being the number of random variables.

In this letter, we discuss the relationship between the Laplace and the variational approximation, and we show that for models with gaussian priors and factorizing likelihoods, the number of variational parameters is actually Omicron(N). The approach is applied to gaussian process regression with nongaussian likelihoods.

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40. Nonnegative Matrix Factorization with the Itakura-Saito Divergence: With Application to Music Analysis

Cédric Févotte, Nancy Bertin, Jean-Louis Durrieu

https://doi.org/10.1162/neco.2008.04-08-771 | pp. 793-830

This letter presents theoretical, algorithmic, and experimental results about nonnegative matrix factorization (NMF) with the Itakura-Saito (IS) divergence. We describe how IS-NMF is underlaid by a well-defined statistical model of superimposed gaussian components and is equivalent to maximum likelihood estimation of variance parameters. This setting can accommodate regularization constraints on the factors through Bayesian priors. In particular, inverse-gamma and gamma Markov chain priors are considered in this work. Estimation can be carried out using a space-alternating generalized expectation-maximization (SAGE) algorithm; this leads to a novel type of NMF algorithm, whose convergence to a stationary point of the IS cost function is guaranteed. We also discuss the links between the IS divergence and other cost functions used in NMF, in particular, the Euclidean distance and the generalized Kullback-Leibler (KL) divergence. As such, we describe how IS-NMF can also be performed using a gradient multiplicative algorithm (a standard algorithm structure in NMF) whose convergence is observed in practice, though not proven. Finally, we report a furnished experimental comparative study of Euclidean-NMF, KL-NMF, and IS-NMF algorithms applied to the power spectrogram of a short piano sequence recorded in real conditions, with various initializations and model orders. Then we show how IS-NMF can successfully be employed for denoising and upmix (mono to stereo conversion) of an original piece of early jazz music. These experiments indicate that IS-NMF correctly captures the semantics of audio and is better suited to the representation of music signals than NMF with the usual Euclidean and KL costs.

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41. Parallel Hopfield Networks

Robert C. Wilson

https://doi.org/10.1162/neco.2008.03-07-496 | pp. 831-850

We introduce a novel type of neural network, termed the parallel Hopfield network, that can simultaneously effect the dynamics of many different, independent Hopfield networks in parallel in the same piece of neural hardware. Numerically we find that under certain conditions, each Hopfield subnetwork has a finite memory capacity approaching that of the equivalent isolated attractor network, while a simple signal-to-noise analysis sheds qualitative, and some quantitative, insight into the workings (and failures) of the system.

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42. Maximum Memory Capacity on Neural Networks with Short-Term Synaptic Depression and Facilitation

Jorge F. Mejías, Joaqu■■n J. Torres

https://doi.org/10.1162/neco.2008.02-08-719 | pp. 851-871

In this work, we study, analytically and employing Monte Carlo simulations, the influence of the competition between several activity-dependent synaptic processes, such as short-term synaptic facilitation and depression, on the maximum memory storage capacity in a neural network. In contrast to the case of synaptic depression, which drastically reduces the capacity of the network to store and retrieve "static" activity patterns, synaptic facilitation enhances the storage capacity in different contexts. In particular, we found optimal values of the relevant synaptic parameters (such as the neurotransmitter release probability or the characteristic facilitation time constant) for which the storage capacity can be maximal and similar to the one obtained with static synapses, that is, without activity-dependent processes. We conclude that depressing synapses with a certain level of facilitation allow recovering the good retrieval properties of networks with static synapses while maintaining the nonlinear characteristics of dynamic synapses, convenient for information processing and coding.

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43. Adaptive Improved Natural Gradient Algorithm for Blind Source Separation

Jianqiang Liu, Da■Zheng Feng, Weiwei Zhang

https://doi.org/10.1162/neco.2008.07-07-562 | pp. 872-889

We propose an adaptive improved natural gradient algorithm for blind separation of independent sources. First, inspired by the well-known backpropagation algorithm, we incorporate a momentum term into the natural gradient learning process to accelerate the convergence rate and improve the stability. Then an estimation function for the adaptation of the separation model is obtained to adaptively control a step-size parameter and a momentum factor. The proposed natural gradient algorithm with variable step-size parameter and variable momentum factor is therefore particularly well suited to blind source separation in a time-varying environment, such as an abruptly changing mixing matrix or signal power. The expected improvement in the convergence speed, stability, and tracking ability of the proposed algorithm is demonstrated by extensive simulation results in both time-invariant and time-varying environments. The ability of the proposed algorithm to separate extremely weak or badly scaled sources is also verified. In addition, simulation results show that the proposed algorithm is suitable for separating mixtures of many sources (e.g., the number of sources is 10) in the complete case.

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44. A Bi-Prototype Theory of Facial Attractiveness

Fu Chang, Chien-Hsing Chou

https://doi.org/10.1162/neco.2008.07-07-566 | pp. 890-910

The attractiveness of human faces can be predicted with a high degree of accuracy if we represent the faces as feature vectors and compute their relative distances from two prototypes: the average of attractive faces and the average of unattractive faces. Moreover, the degree of attractiveness, defined in terms of the relative distance, exhibits a high degree of correlation with the average rating scores given by human assessors. These findings motivate a bi-prototype theory that relates facial attractiveness to the averages of attractive and unattractive faces rather than the average of all faces, as previously hypothesized by some researchers.

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45. Spiking Neurons Can Learn to Solve Information Bottleneck Problems and Extract Independent Components

Stefan Klampfl, Robert Legenstein, Wolfgang Maass

https://doi.org/10.1162/neco.2008.01-07-432 | pp. 911-959

Independent component analysis (or blind source separation) is assumed to be an essential component of sensory processing in the brain and could provide a less redundant representation about the external world. Another powerful processing strategy is the optimization of internal representations according to the information bottleneck method. This method would allow extracting preferentially those components from high-dimensional sensory input streams that are related to other information sources, such as internal predictions or proprioceptive feedback. However, there exists a lack of models that could explain how spiking neurons could learn to execute either of these two processing strategies. We show in this article how stochastically spiking neurons with refractoriness could in principle learn in an unsupervised manner to carry out both information bottleneck optimization and the extraction of independent components. We derive suitable learning rules, which extend the well-known BCM rule, from abstract information optimization principles. These rules will simultaneously keep the firing rate of the neuron within a biologically realistic range.

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46. Measure of Correlation Orthogonal to Change in Firing Rate

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https://doi.org/10.1162/neco.2008.03-08-729 | pp. 960-972

There are a number of measures of correlation for spikes of two neurons and for spikes at two successive time bins in one neuron: covariance, correlation coefficient, mutual information, and information-geometric measure in the log-linear model. It is desirable to have a measure that is not affected by change in the firing rates of neurons. We explain the superiority of the information-geometric measure from the point of view of geometry, by which the correlation and firing rates are separated orthogonally, that is, without correlation. We then analyze characteristics of other measures and show analytically how they are related to firing rates.

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47. Wide-Band Information Transmission at the Calyx of Held

Zhijun Yang, Matthias H. Hennig, Michael Postlethwaite, Ian D. Forsythe, Bruce Graham

https://doi.org/10.1162/neco.2008.02-08-714 | pp. 991-1017

We use a mathematical model of the calyx of Held to explore information transmission at this giant glutamatergic synapse. The significant depression of the postsynaptic response to repeated stimulation in vitro is a result of various activity-dependent processes in multiple timescales, which can be reproduced by multiexponential functions in this model. When the postsynaptic current is stimulated by Poisson-distributed spike trains, its amplitude varies considerably with the preceding interspike intervals. Here we quantify the information contained in the postsynaptic current amplitude about preceding interspike intervals and determine the impact of different pre- and postsynaptic factors on information transmission. The mutual information between presynaptic spike times and the amplitude of the postsynaptic response in general decreases as the mean stimulation rate increases, but remains high even at frequencies greater than 100 Hz, unlike at many neocortical synapses. The maintenance of information transmission is attributable largely

to vesicle recycling rates at low frequencies of stimulation, shifting to vesicle release probability at high frequencies. Also, at higher frequencies, the synapse operates largely in a release-ready mode in which most release sites contain a release-ready vesicle and release probabilities are low.

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48. Chemosensor-Driven Artificial Antennal Lobe Transient Dynamics Enable Fast Recognition and Working Memory

Mehmet K. Muezzinoglu, Ramón Huerta, Henry D. I. Abarbanel, M. A. Ryan, M. I. Rabinovich
https://doi.org/10.1162/neco.2008.05-08-780 | pp. 1018-1037

The speed and accuracy of odor recognition in insects can hardly be resolved by the raw descriptors provided by olfactory receptors alone due to their slow time constant and high variability. The animal overcomes these barriers by means of the antennal lobe (AL) dynamics, which consolidates the classificatory information in receptor signal with a spatiotemporal code that is enriched in odor sensitivity, particularly in its transient. Inspired by this fact, we propose an easily implementable AL-like network and show that it significantly expedites and enhances the identification of odors from slow and noisy artificial polymer sensor responses. The device owes its efficiency to two intrinsic mechanisms: inhibition (which triggers a competition) and integration (due to the dynamical nature of the network). The former functions as a sharpening filter extracting the features of receptor signal that favor odor separation, whereas the latter implements a working memory by accumulating the extracted features in trajectories. This cooperation boosts the odor specificity during the receptor transient, which is essential for fast odor recognition.

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49. Recurrent Infomax Generates Cell Assemblies, Neuronal Avalanches, and Simple Cell-Like Selectivity

Takuma Tanaka, Takeshi Kaneko, Toshio Aoyagi
https://doi.org/10.1162/neco.2008.03-08-727 | pp. 1038-1067

Recently multineuronal recording has allowed us to observe patterned firings, synchronization, oscillation, and global state transitions in the recurrent networks of central nervous systems. We propose a learning algorithm based on the process of information maximization in a recurrent network, which we call recurrent infomax (RI). RI maximizes information retention and thereby minimizes information loss through time in a network. We find that feeding in external inputs consisting of information obtained from photographs of natural scenes into an RI-based model of a recurrent network results in the appearance of Gabor-like selectivity quite similar to that existing in simple cells of the primary visual cortex. We find that without external input, this network exhibits cell assembly-like and synfire chain-like spontaneous activity as well as a critical neuronal avalanche. In addition, we find that RI embeds externally input temporal firing patterns to the network so that it spontaneously reproduces these patterns after learning. RI provides a simple framework to explain a wide range of phenomena observed in in vivo and in vitro neuronal networks, and it will provide a novel understanding of experimental results for multineuronal activity and plasticity from an information-theoretic point of view.

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50. Accelerating Event-Driven Simulation of Spiking Neurons with Multiple Synaptic Time Constants

Michiel D'Haene, Benjamin Schrauwen, Jan Van Campenhout, Dirk Stroobandt

https://doi.org/10.1162/neco.2008.02-08-707 | pp. 1068-1099

The simulation of spiking neural networks (SNNs) is known to be a very time-consuming task. This limits the size of SNN that can be simulated in reasonable time or forces users to overly limit the complexity of the neuron models. This is one of the driving forces behind much of the recent research on event-driven simulation strategies. Although event-driven simulation allows precise and efficient simulation of certain spiking neuron models, it is not straightforward to generalize the technique to more complex neuron models, mostly because the firing time of these neuron models is computationally expensive to evaluate. Most solutions proposed in literature concentrate on algorithms that can solve this problem efficiently. However, these solutions do not scale well when more state variables are involved in the neuron model, which is, for example, the case when multiple synaptic time constants for each neuron are used. In this letter, we show that an exact prediction of the firing time is not required in order to guarantee exact simulation results. Several techniques are presented that try to do the least possible amount of work to predict the firing times. We propose an elegant algorithm for the simulation of leaky integrate-and-fire (LIF) neurons with an arbitrary number of (unconstrained) synaptic time constants, which is able to combine these algorithmic techniques efficiently, resulting in very high simulation speed. Moreover, our algorithm is highly independent of the complexity (i.e., number of synaptic time constants) of the underlying neuron model.

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51. Delays and Oscillations in Networks of Spiking Neurons: A Two-Timescale Analysis

Dotan Di Castro, Ron Meir, Irad Yavneh

https://doi.org/10.1162/neco.2008.03-08-723 | pp. 1100-1124

Oscillations are a ubiquitous feature of many neural systems, spanning many orders of magnitude in frequency. One of the most prominent oscillatory patterns, with possible functional implications, is that occurring in the mammalian thalamocortical system during sleep. This system is characterized by relatively long delays (reaching up to 40 msec) and gives rise to low-frequency oscillatory waves. Motivated by these phenomena, we study networks of excitatory and inhibitory integrate-and-fire neurons within a Fokker-Planck delay partial differential equation formalism and establish explicit conditions for the emergence of oscillatory solutions, and for the amplitude and period of the ensuing oscillations, for relatively large values of the delays. When a two-timescale analysis is employed, the full partial differential equation is replaced in this limit by a discrete time iterative map, leading to a relatively simple dynamic interpretation. This asymptotic result is shown numerically to hold, to a good approximation, over a wide range of parameter values, leading to an accurate characterization of the behavior in terms of the underlying physical parameters. Our results provide a simple mechanistic explanation for one type of slow oscillation based on delayed inhibition, which may play an important role in the slow spindle oscillations occurring during sleep. Moreover, they are consistent with experimental findings related to human motor behavior with visual feedback.

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52. Searching for Memories, Sudoku, Implicit Check Bits, and the Iterative Use of Not-Always-Correct Rapid Neural Computation

J. J. Hopfield

https://doi.org/10.1162/neco.2007.09-06-345 | pp. 1119-1164

The algorithms that simple feedback neural circuits representing a brain area can rapidly carry out are often adequate to solve easy problems but for more difficult problems can return incorrect answers. A new excitatory-inhibitory circuit model of associative memory displays the common human problem of failing to rapidly find a memory when only a small clue is present. The memory model and a related computational network for solving Sudoku puzzles produce answers that contain implicit check bits in the representation of information across neurons, allowing a rapid evaluation of whether the putative answer is correct or incorrect through a computation related to visual pop-out. This fact may account for our strong psychological feeling of right or wrong when we retrieve a nominal memory from a minimal clue. This information allows more difficult computations or memory retrievals to be done in a serial fashion by using the fast but limited capabilities of a computational module multiple times. The mathematics of the excitatory-inhibitory circuits for associative memory and for Sudoku, both of which are understood in terms of energy or Lyapunov functions, is described in detail.

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54. Unsupervised Learning of Individuals and Categories from Images

S. Waydo, Christof Koch

https://doi.org/10.1162/neco.2007.03-07-493 | pp. 1165-1178

Motivated by the existence of highly selective, sparsely firing cells observed in the human medial temporal lobe (MTL), we present an unsupervised method for learning and recognizing object categories from unlabeled images. In our model, a network of nonlinear neurons learns a sparse representation of its inputs

through an unsupervised expectation-maximization process. We show that the application of this strategy to an invariant feature-based description of natural images leads to the development of units displaying sparse, invariant selectivity for particular individuals or image categories much like those observed in the MTL data.

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56. On the Asymptotic Equivalence Between Differential Hebbian and Temporal Difference Learning

Christoph Kolodziejski, Bernd Porr, Florentin Wörgötter

https://doi.org/10.1162/neco.2008.04-08-750 | pp. 1173-1202

In this theoretical contribution, we provide mathematical proof that two of the most important classes of network learning-correlation-based differential Hebbian learning and reward-based temporal difference learning-are asymptotically equivalent when timing the learning with a modulatory signal. This opens the opportunity to consistently reformulate most of the abstract reinforcement learning framework from a correlation-based perspective more closely related to the biophysics of neurons.

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57. Populations of Tightly Coupled Neurons: The RGC/LGN System

Lawrence Sirovich

https://doi.org/10.1162/neco.2008.03-07-482 | pp. 1179-1210

A mathematical model, of general character for the dynamic description of coupled neural oscillators is presented. The population approach that is employed applies equally to coupled cells as to populations of such coupled cells. The formulation includes stochasticity and preserves details of precisely firing neurons. Based on the generally accepted view of cortical wiring, this formulation is applied to the retinal ganglion cell (RGC)/lateral geniculate nucleus (LGN) relay cell system, of the early mammalian visual system. The smallness of quantal voltage jumps at the retinal level permits a Fokker-Planck approximation for the RGC contribution; however, the LGN description requires the use of finite jumps, which for fast synaptic dynamics appears as finite jumps in the membrane potential.Analyses of equilibrium spiking behavior for both the deterministic and stochastic cases are presented. Green's function methods form the basis for the asymptotic and exact results that are presented. This determines the spiking ratio (i.e., the number of RGC arrivals per LGN spike), which is the reciprocal of the transfer ratio, under wide circumstances. Criteria for spiking regimes, in terms of the relatively few parameters of the model, are presented.Under reasonable

hypotheses, it is shown that the transfer ratio is ≤1/2, in the absence of input from other areas. Thus, the model suggests that the LGN/RGC system may be a relatively unsophisticated spike editor. In the absence of other input, the system is designed to fire an LGN spike only when two or more RGC spikes appear in a relatively short time. Transfer ratios that briefly exceed 1/2 (but are less than 1) have been recorded in the laboratory. Inclusion of brain stem input has been shown to provide a signal that elevates the transfer ratio (Ozaki & Kaplan, 2006). A model that includes this contribution is also presented.

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58. Messages of Oscillatory Correlograms: A Spike Train Model

Gaby Schneider

https://doi.org/10.1162/neco.2007.12-06-424 | pp. 1211-1238

Oscillatory correlograms are widely used to study neuronal activity that shows a joint periodic rhythm. In most cases, the statistical analysis of cross-correlation histograms (CCH) features is based on the null model of independent processes, and the resulting conclusions about the underlying processes remain qualitative. Therefore, we propose a spike train model for synchronous oscillatory firing activity that directly links characteristics of the CCH to parameters of the underlying processes. The model focuses particularly on asymmetric central peaks, which differ in slope and width on the two sides. Asymmetric peaks can be associated with phase offsets in the (sub-) millisecond range. These spatiotemporal firing patterns can be highly consistent across units yet invisible in the underlying processes. The proposed model includes a single temporal parameter that accounts for this peak asymmetry. The model provides approaches for the analysis of oscillatory correlograms, taking into account dependencies and nonstationarities in the underlying processes. In particular, the auto- and the cross-correlogram can be investigated in a joint analysis because they depend on the same spike train parameters. Particular temporal interactions such as the degree to which different units synchronize in a common oscillatory rhythm can also be investigated. The analysis is demonstrated by application to a simulated data set.

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60. Two Computational Regimes of a Single-Compartment Neuron Separated by a Planar Boundary in Conductance Space

Brian N. Lundstrom, Sungho Hong, Matthew H. Higgs, Adrienne L. Fairhall

https://doi.org/10.1162/neco.2007.05-07-536 | pp. 1239-1260

Recent in vitro data show that neurons respond to input variance with varying sensitivities. Here we demonstrate that Hodgkin-Huxley (HH) neurons can operate in two computational regimes: one that is more sensitive to input variance (differentiating) and one that is less sensitive (integrating). A boundary plane in the 3D conductance space separates these two regimes. For a reduced HH model, this plane can be derived analytically from the V nullcline, thus suggesting a means of relating biophysical parameters to neural computation by analyzing the neuron's dynamical system.

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62. A Rate and History-Preserving Resampling Algorithm for Neural Spike Trains

Matthew Tom Harrison, Stuart Geman

https://doi.org/10.1162/neco.2008.03-08-730 | pp. 1244-1258

Resampling methods are popular tools for exploring the statistical structure of neural spike trains. In many applications, it is desirable to have resamples that preserve certain non-Poisson properties, like refractory periods and bursting, and that are also robust to trial-to-trial variability. Pattern jitter is a resampling technique that accomplishes this by preserving the recent spiking history of all spikes and constraining resampled spikes to remain close to their original positions. The resampled spike times are maximally random up to these constraints. Dynamic programming is used to create an efficient resampling algorithm.

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63. A Sparse Generative Model of V1 Simple Cells with Intrinsic Plasticity

Cornelius Weber, Jochen Triesch

https://doi.org/10.1162/neco.2007.02-07-472 | pp. 1261-1284

Current models for learning feature detectors work on two timescales: on a fast timescale, the internal neurons' activations adapt to the current stimulus; on a slow timescale, the weights adapt to the statistics of the set of stimuli. Here we explore the adaptation of a neuron's intrinsic excitability, termed intrinsic plasticity, which occurs on a separate timescale. Here, a neuron maintains homeostasis of an exponentially distributed firing rate in a dynamic environment. We exploit this in the context of a generative model to impose sparse coding. With natural image input, localized edge detectors emerge as models of V1 simple cells. An intermediate timescale for the intrinsic plasticity parameters allows modeling aftereffects. In the tilt aftereffect, after a viewer adapts to a grid of a certain orientation, grids of a nearby orientation will be perceived as tilted away from the adapted orientation. Our results show that adapting the neurons' gain-parameter but not the threshold-parameter accounts for this effect. It occurs because neurons coding for the adapting stimulus attenuate their gain, while others increase it. Despite its simplicity and low maintenance, the intrinsic plasticity model accounts for more experimental details than previous models without this mechanism.

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65. Bayesian Filtering in Spiking Neural Networks: Noise, Adaptation, and Multisensory Integration

Omer Bobrowski, Ron Meir, Yonina C. Eldar

https://doi.org/10.1162/neco.2008.01-08-692 | pp. 1277-1320

A key requirement facing organisms acting in uncertain dynamic environments is the real-time estimation and prediction of environmental states, based on which effective actions can be selected. While it is becoming evident that organisms employ exact or approximate Bayesian statistical calculations for these purposes, it is far less clear how these putative computations are implemented by neural networks in a

strictly dynamic setting. In this work, we make use of rigorous mathematical results from the theory of continuous time point process filtering and show how optimal real-time state estimation and prediction may be implemented in a general setting using simple recurrent neural networks. The framework is applicable to many situations of common interest, including noisy observations, non-Poisson spike trains (incorporating adaptation), multisensory integration, and state prediction. The optimal network properties are shown to relate to the statistical structure of the environment, and the benefits of adaptation are studied and explicitly demonstrated. Finally, we recover several existing results as appropriate limits of our general setting.

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66. Parameters of Spike Trains Observed in a Short Time Window

Zbyn■k Pawlas, Lev B. Klebanov, Martin Prokop, Petr Lánský

https://doi.org/10.1162/neco.2007.01-07-442 | pp. 1325-1343

We study the estimation of statistical moments of interspike intervals based on observation of spike counts in many independent short time windows. This scenario corresponds to the situation in which a target neuron occurs. It receives information from many neurons and has to respond within a short time interval. The precision of the estimation procedures is examined. As the model for neuronal activity, two examples of stationary point processes are considered: renewal process and doubly stochastic Poisson process. Both moment and maximum likelihood estimators are investigated. Not only the mean but also the coefficient of variation is estimated. In accordance with our expectations, numerical studies confirm that the estimation of mean interspike interval is more reliable than the estimation of coefficient of variation. The error of estimation increases with increasing mean interspike interval, which is equivalent to decreasing the size of window (less events are observed in a window) and with decreasing the number of neurons (lower number of windows).

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68. A Computational Model for Rhythmic and Discrete Movements in Uni- and Bimanual Coordination

Renaud Ronsse, Dagmar Sternad, Philippe Lefèvre

https://doi.org/10.1162/neco.2008.03-08-720 | pp. 1335-1370

Current research on discrete and rhythmic movements differs in both experimental procedures and theory, despite the ubiquitous overlap between discrete and rhythmic components in everyday behaviors. Models of rhythmic movements usually use oscillatory systems mimicking central pattern generators (CPGs). In contrast, models of discrete movements often employ optimization principles, thereby reflecting the higher-level cortical resources involved in the generation of such movements. This letter proposes a unified model for the generation of both rhythmic and discrete movements. We show that a physiologically motivated model of a CPG can not only generate simple rhythmic movements with only a small set of parameters, but can also produce discrete movements if the CPG is fed with an exponentially decaying phasic input. We further show that a particular coupling between two of these units can reproduce main findings on in-phase and antiphase stability. Finally, we propose an integrated model of combined rhythmic and discrete movements for the two hands. These movement classes are sequentially addressed in this letter with increasing model complexity. The model variations are discussed in relation to the degree of recruitment of the higher-level cortical resources, necessary for such movements.

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69. A One-Layer Recurrent Neural Network with a Discontinuous Activation Function for Linear Programming

Qingshan Liu, Jun Wang

https://doi.org/10.1162/neco.2008.03-07-488 | pp. 1366-1383

A one-layer recurrent neural network with a discontinuous activation function is proposed for linear programming. The number of neurons in the neural network is equal to that of decision variables in the linear programming problem. It is proven that the neural network with a sufficiently high gain is globally convergent to the optimal solution. Its application to linear assignment is discussed to demonstrate the utility of the neural network. Several simulation examples are given to show the effectiveness and characteristics of the neural network.

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70. Analysis of the Interaction Between the Dendritic Conductance Density and Activated Area in Modulating α-Motoneuron EPSP: Statistical Computer Model

Gideon Gradwohl, Yoram Grossman

https://doi.org/10.1162/neco.2008.03-07-490 | pp. 1385-1410

Five reconstructed alpha-motoneurons (MNs) are simulated under physiological and morphological realistic parameters. We compare the resulting excitatory postsynaptic potential (EPSP) of models, containing voltage-dependent channels on the dendrites, with the EPSP of a passive MN and an active soma and axon model. In our simulations, we apply three different distribution functions of the voltage-dependent channels on the dendrites: a step function (ST) with uniform spatial dispersion; an exponential decay (ED) function, with proximal to the soma high-density location; and an exponential rise (ER) with distally located

conductance density. In all cases, the synaptic inputs are located as a gaussian function on the dendrites. Our simulations lead to eight key observations. (1) The presence of the voltage-dependent channels conductance (g(Active)) in the dendrites is vital for obtaining EPSP peak boosting. (2) The mean EPSP peaks of the ST, ER, and ED distributions are similar when the ranges of G (total conductance) are equal. (3) EPSP peak increases monotonically when the magnitude of g(Na_step) (maximal g(Na) at a particular run) is increased. (4) EPSP kinetics parameters were differentially affected; time integral was decreased monotonically with increased g(Na_step), but the rate of rise (the decay time was not analyzed) does not show clear relations. (5) The total G can be elevated by increasing the number of active dendrites; however, only a small active area of the dendritic tree is sufficient to get the maximal boosting. (6) The sometimes large variations in the parameters values for identical G depend on the g(Na_step) and active dendritic area. (7) High g(Na_step) in a few dendrites is more efficient in amplifying the EPSP peak than low g(Na_step) in many dendrites. (8) The EPSP peak is approximately linear with respect to the MNs' R(N) (input resistance).

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71. Discrimination with Spike Times and ISI Distributions

Kukjin Kang, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco.2007.07-07-561 | pp. 1411-1426

We study the discrimination capability of spike time sequences using the Chernoff distance as a metric. We assume that spike sequences are generated by renewal processes and study how the Chernoff distance depends on the shape of interspike interval (ISI) distribution. First, we consider a lower bound to the Chernoff distance because it has a simple closed form. Then we consider specific models of ISI distributions such as the gamma, inverse gaussian (IG), exponential with refractory period (ER), and that of the leaky integrate-and-fire (LIF) neuron. We found that the discrimination capability of spike times strongly depends on high-order moments of ISI and that it is higher when the spike time sequence has a larger skewness and a smaller kurtosis. High variability in terms of coefficient of variation (CV) does not necessarily mean that the spike times have less discrimination capability. Spike sequences generated by the gamma distribution have the minimum discrimination capability for a given mean and variance of ISI. We used series expansions to calculate the mean and variance of ISIs for LIF neurons as a function of the mean input level and the input noise variance. Spike sequences from an LIF neuron are more capable of discrimination than those of IG and gamma distributions when the stationary voltage level is close to the neuron's threshold value of the neuron.

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72. A Geometric Newton Method for Oja's Vector Field

P.-A. Absil, M. Ishteva, L. De Lathauwer, S. Van Huffel

https://doi.org/10.1162/neco.2008.04-08-749 | pp. 1415-1433

Newton's method for solving the matrix equation F(X) identical to AX-XX(T) AX = 0 runs up against the fact that its zeros are not isolated. This is due to a symmetry of F by the action of the orthogonal group. We show how differential-geometric techniques can be exploited to remove this symmetry and obtain a "geometric" Newton algorithm that finds the zeros of F. The geometric Newton method does not suffer from the degeneracy issue that stands in the way of the original Newton method.

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73. A Canonical Neural Circuit for Cortical Nonlinear Operations

Minjoon Kouh, Tomaso Poggio

https://doi.org/10.1162/neco.2008.02-07-466 | pp. 1427-1451

A few distinct cortical operations have been postulated over the past few years, suggested by experimental data on nonlinear neural response across different areas in the cortex. Among these, the energy model proposes the summation of quadrature pairs following a squaring nonlinearity in order to explain phase invariance of complex V1 cells. The divisive normalization model assumes a gain-controlling, divisive inhibition to explain sigmoid-like response profiles within a pool of neurons. A gaussian-like operation hypothesizes a bell-shaped response tuned to a specific, optimal pattern of activation of the presynaptic inputs. A max-like operation assumes the selection and transmission of the most active response among a set of neural inputs. We propose that these distinct neural operations can be computed by the same canonical circuitry, involving divisive normalization and polynomial nonlinearities, for different parameter values within the circuit. Hence, this canonical circuit may provide a unifying framework for several circuit models, such as the divisive normalization and the energy models. As a case in point, we consider a feedforward hierarchical model of the ventral pathway of the primate visual cortex, which is built on a combination of the gaussian-like and max-like operations. We show that when the two operations are approximated by the circuit proposed here, the model is capable of generating selective and invariant neural responses and performing object recognition, in good agreement with neurophysiological data.

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74. Figure-Ground Separation by Cue Integration

Xiangyu Tang, Christoph von der Malsburg

https://doi.org/10.1162/neco.2008.03-06-176 | pp. 1452-1472

This letter presents an improved cue integration approach to reliably separate coherent moving objects from their background scene in video sequences. The proposed method uses a probabilistic framework to unify bottom-up and top-down cues in a parallel, "democratic" fashion. The algorithm makes use of a modified Bayes rule where each pixel's posterior probabilities of figure or ground layer assignment are derived from likelihood models of three bottom-up cues and a prior model provided by a top-down cue. Each cue is treated as independent evidence for figure-ground separation. They compete with and complement each other dynamically by adjusting relative weights from frame to frame according to cue quality measured against the overall integration. At the same time, the likelihood or prior models of individual cues adapt toward the integrated result. These mechanisms enable the system to organize under the influence of visual scene structure without manual intervention. A novel contribution here is the incorporation of a top-down cue. It improves the system's robustness and accuracy and helps handle difficult and ambiguous situations, such as abrupt lighting changes or occlusion among multiple objects. Results on various video sequences are demonstrated and discussed. (Video demos are available at http://organic.usc.edu:8376/ approximately tangx/neco/index.html .).

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75. Solving the Problem of Negative Synaptic Weights in Cortical Models

Christopher Parisien, Charles H. Anderson, Chris Eliasmith

https://doi.org/10.1162/neco.2008.07-06-295 | pp. 1473-1494

In cortical neural networks, connections from a given neuron are either inhibitory or excitatory but not both. This constraint is often ignored by theoreticians who build models of these systems. There is currently no general solution to the problem of converting such unrealistic network models into biologically plausible models that respect this constraint. We demonstrate a constructive transformation of models that solves this problem for both feedforward and dynamic recurrent networks. The resulting models give a close approximation to the original network functions and temporal dynamics of the system, and they are biologically plausible. More precisely, we identify a general form for the solution to this problem. As a result, we also describe how the precise solution for a given cortical network can be determined empirically.

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76. A New Multineuron Spike Train Metric

Conor Houghton, Kamal Sen

https://doi.org/10.1162/neco.2007.10-06-350 | pp. 1495-1511

The Victor-Purpura spike train metric has recently been extended to a family of multineuron metrics and used to analyze spike trains recorded simultaneously from pairs of proximate neurons. The metric is one of the two metrics commonly used for quantifying the distance between two spike trains; the other is the van Rossum metric. Here, we suggest an extension of the van Rossum metric to a multineuron metric. We believe this gives a metric that is both natural and easy to calculate. Both types of multineuron metric are applied to simulated data and are compared.

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77. A Classification Method to Distinguish Cell-Specific Responses Elicited by Current Pulses in Hippocampal CA1 Pyramidal Cells

José Ambros■Ingerson, Lawrence M. Grover, William R. Holmes

https://doi.org/10.1162/neco.2007.07-07-564 | pp. 1512-1536

The suprathreshold electrophysiological responses of pyramidal cells have been grouped into large classes such as bursting and spiking. However, it is not known whether, within a class, response variability ranges uniformly across all cells or whether each cell has a unique and consistent profile that can be differentiated. A major difficulty when comparing suprathreshold responses is that slight variations in spike timing in otherwise very similar traces render traditional metrics ineffective. To address these issues, we developed a novel distance measure based on fiducial points to quantify the similarity among traces with trains of action potentials and applied it together with classification techniques to a set of in vitro patch clamp recordings from CA1 pyramidal cells. We tested if responses to repeated current stimulation of a given cell would cluster together yet remain distinct from those of other cells. We found that depolarizing and hyperpolarizing current pulses elicited responses in each cell that clustered and were systematically distinguishable from responses in other cells. The fiducial-point distance measure was more effective than other methods based on spike times and voltage-gradient phase planes. Depolarizing traces were more reliably differentiated than hyperpolarizing traces, and combining both scores was even more effective. These results suggest that each CA1 pyramidal cell has unique properties that can be detected and quantified with methods discussed here. This uniqueness may be due to slight variations in morphology or membrane channel densities and kinetics, or to large, coordinated variations in these elements. Ascertaining the actual sources and their degree of variability is important when constructing network models of neural function to ensure that key mechanisms are robust in the face of variations within these ranges. The analytical tools presented here can

assist in constructing detailed cell models to match experimental records to elucidate the sources of electrophysiological variability in neurons.

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78. The Berkeley Wavelet Transform: A Biologically Inspired Orthogonal Wavelet Transform

Ben D. B. Willmore, Ryan Prenger, Michael C. Wu, Jack L. Gallant

https://doi.org/10.1162/neco.2007.05-07-513 | pp. 1537-1564

We describe the Berkeley wavelet transform (BWT), a two-dimensional triadic wavelet transform. The BWT comprises four pairs of mother wavelets at four orientations. Within each pair, one wavelet has odd symmetry, and the other has even symmetry. By translation and scaling of the whole set (plus a single constant term), the wavelets form a complete, orthonormal basis in two dimensions. The BWT shares many characteristics with the receptive fields of neurons in mammalian primary visual cortex (V1). Like these receptive fields, BWT wavelets are localized in space, tuned in spatial frequency and orientation, and form a set that is approximately scale invariant. The wavelets also have spatial frequency and orientation bandwidths that are comparable with biological values. Although the classical Gabor wavelet model is a more accurate description of the receptive fields of individual V1 neurons, the BWT has some interesting advantages. It is a complete, orthonormal basis and is therefore inexpensive to compute, manipulate, and invert. These properties make the BWT useful in situations where computational power or experimental data are limited, such as estimation of the spatiotemporal receptive fields of neurons.

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79. Deterministic Neural Classification

Kar■Ann Toh

https://doi.org/10.1162/neco.2007.04-07-508 | pp. 1565-1595

This letter presents a minimum classification error learning formulation for a single-layer feedforward network (SLFN). By approximating the nonlinear counting step function using a quadratic function, the classification error rate is shown to be deterministically solvable. Essentially the derived solution is related to an existing weighted least-squares method with class-specific weights set according to the size of data set. By considering the class-specific weights as adjustable parameters, the learning formulation extends the classification robustness of the SLFN without sacrificing its intrinsic advantage of being a closed-form algorithm. While the method is applicable to other linear formulations, our empirical results indicate SLFN's effectiveness on classification generalization.

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80. Robust Boosting Algorithm Against Mislabeling in Multiclass Problems

Takashi Takenouchi, Shinto Eguchi, Noboru Murata, Takafumi Kanamori

https://doi.org/10.1162/neco.2007.11-06-400 | pp. 1596-1630

We discuss robustness against mislabeling in multiclass labels for classification problems and propose two algorithms of boosting, the normalized Eta-Boost.M and Eta-Boost.M, based on the Eta-divergence. Those two boosting algorithms are closely related to models of mislabeling in which the label is erroneously exchanged for others. For the two boosting algorithms, theoretical aspects supporting the robustness for

mislabeling are explored. We apply the proposed two boosting methods for synthetic and real data sets to investigate the performance of these methods, focusing on robustness, and confirm the validity of the proposed methods.

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81. Justifying and Generalizing Contrastive Divergence

Yoshua Bengio, Olivier Delalleau

https://doi.org/10.1162/neco.2008.11-07-647 | pp. 1601-1621

We study an expansion of the log likelihood in undirected graphical models such as the restricted Boltzmann machine (RBM), where each term in the expansion is associated with a sample in a Gibbs chain alternating between two random variables (the visible vector and the hidden vector in RBMs). We are particularly interested in estimators of the gradient of the log likelihood obtained through this expansion. We show that its residual term converges to zero, justifying the use of a truncation--running only a short Gibbs chain, which is the main idea behind the contrastive divergence (CD) estimator of the log-likelihood gradient. By truncating even more, we obtain a stochastic reconstruction error, related through a mean-field approximation to the reconstruction error often used to train autoassociators and stacked autoassociators. The derivation is not specific to the particular parametric forms used in RBMs and requires only convergence of the Gibbs chain. We present theoretical and empirical evidence linking the number of Gibbs steps k and the magnitude of the RBM parameters to the bias in the CD estimator. These experiments also suggest that the sign of the CD estimator is correct most of the time, even when the bias is large, so that CD-k is a good descent direction even for small k.

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82. Representational Power of Restricted Boltzmann Machines and Deep Belief Networks

Nicolas Le Roux, Yoshua Bengio

https://doi.org/10.1162/neco.2008.04-07-510 | pp. 1631-1649

Deep belief networks (DBN) are generative neural network models with many layers of hidden explanatory factors, recently introduced by Hinton, Osindero, and Teh (2006) along with a greedy layer-wise unsupervised learning algorithm. The building block of a DBN is a probabilistic model called a restricted Boltzmann machine (RBM), used to represent one layer of the model. Restricted Boltzmann machines are interesting because inference is easy in them and because they have been successfully used as building blocks for training deeper models. We first prove that adding hidden units yields strictly improved modeling power, while a second theorem shows that RBMs are universal approximators of discrete distributions. We then study the question of whether DBNs with more layers are strictly more powerful in terms of representational power. This suggests a new and less greedy criterion for training RBMs within DBNs.

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83. Theory of Input Spike Auto- and Cross-Correlations and Their Effect on the Response of Spiking Neurons

Rubén Moreno■Bote, Alfonso Renart, Néstor Parga

https://doi.org/10.1162/neco.2008.03-07-497 | pp. 1651-1705

Spike correlations between neurons are ubiquitous in the cortex, but their role is not understood. Here we describe the firing response of a leaky integrate-and-fire neuron (LIF) when it receives a temporarily correlated input generated by presynaptic correlated neuronal populations. Input correlations are characterized in terms of the firing rates, Fano factors, correlation coefficients, and correlation timescale of the neurons driving the target neuron. We show that the sum of the presynaptic spike trains cannot be well described by a Poisson process. In fact, the total input current has a nontrivial two-point correlation function described by two main parameters: the correlation timescale (how precise the input correlations are in time) and the correlation magnitude (how strong they are). Therefore, the total current generated by the input spike trains is not well described by a white noise gaussian process. Instead, we model the total current as a colored gaussian process with the same mean and two-point correlation function, leading to the formulation of the problem in terms of a Fokker-Planck equation. Solutions of the output firing rate are found in the limit of short and long correlation timescales. The solutions described here expand and improve on our previous results (Moreno, de la Rocha, Renart, & Parga, 2002) by presenting new analytical expressions for the output firing rate for general IF neurons, extending the validity of the results for arbitrarily large correlation magnitude, and by describing the differential effect of correlations on the mean-driven or noise-dominated firing regimes. Also the details of this novel formalism are given here for the first time. We employ numerical simulations to confirm the analytical solutions and study the firing response to sudden changes in the input correlations. We expect this formalism to be useful for the study of correlations in neuronal networks and their role in neural processing and information transmission.

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84. Online Learning with Hidden Markov Models

Gianluigi Mongillo, Sophie Denève

https://doi.org/10.1162/neco.2008.10-06-351 | pp. 1706-1716

We present an online version of the expectation-maximization (EM) algorithm for hidden Markov models (HMMs). The sufficient statistics required for parameters estimation is computed recursively with time, that is, in an online way instead of using the batch forward-backward procedure. This computational scheme is generalized to the case where the model parameters can change with time by introducing a discount factor into the recurrence relations. The resulting algorithm is equivalent to the batch EM algorithm, for appropriate discount factor and scheduling of parameters update. On the other hand, the online algorithm is able to deal with dynamic environments, i.e., when the statistics of the observed data is changing with time. The implications of the online algorithm for probabilistic modeling in neuroscience are briefly discussed.

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85. Effects of Synaptic Synchrony on the Neuronal Input-Output Relationship

Xiaoshen Li, Giorgio A. Ascoli

https://doi.org/10.1162/neco.2008.10-06-385 | pp. 1717-1731

The firing rate of individual neurons depends on the firing frequency of their distributed synaptic inputs, with linear and nonlinear relations subserving different computational functions. This letter explores the relationship between the degree of synchrony among excitatory synapses and the linearity of the response using detailed compartmental models of cortical pyramidal cells. Synchronous input resulted in a linear input-output relationship, while asynchronous stimulation yielded sub- and supraproportional outputs at low and high frequencies, respectively. The dependence of input-output linearity on synchrony was sigmoidal

and considerably robust with respect to dendritic location, stimulus irregularity, and alteration of active and synaptic properties. Moreover, synchrony affected firing rate differently at lower and higher input frequencies. A reduced integrate-and-fire model suggested a mechanism explaining these results based on spatiotemporal integration, with fundamental implications relating synchrony to memory encoding.

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86. Extending Cable Theory to Heterogeneous Dendrites

Claude Meunier, Boris Lamotte d'Incamps

https://doi.org/10.1162/neco.2008.12-06-425 | pp. 1732-1775

Dendrites may exhibit many types of electrical and morphological heterogeneities at the scale of a few micrometers. Models of neurons, even so-called detailed models, rarely consider such heterogeneities. Small-scale fluctuations in the membrane conductances and the diameter of dendrites are generally disregarded and spines merely incorporated into the dendritic shaft. Using the two-scales method known as homogenization, we establish explicit expressions for the small-scale fluctuations of the membrane voltage, and we derive the cable equation satisfied by the voltage when these fluctuations are averaged out. This allows us to rigorously establish under what conditions a heterogeneous dendrite can be approximated by a homogeneous cable. We consider different distributions of synapses, orderly or random, on a passive dendrite, and we investigate when replacing excitatory and inhibitory synaptic conductances by their local averages leads to a small error in the voltage. This indicates in which regimes the approximations made in compartmental models are justified. We extend these results to active membranes endowed with voltage-dependent conductances or NMDA receptors. Then we examine under which conditions a spiny dendrite behaves as a smooth dendrite. We discover a new regime where this holds true, namely, when the conductance of the spine neck is small compared to the conductance of the synapses impinging on the spine head. Spines can then be taken into account by an effective excitatory current, the capacitance of the dendrite remaining unchanged. In this regime, the synaptic current transmitted from a spine to the dendritic shaft is strongly attenuated by the weak coupling conductance, but the total current they deliver can be quite substantial. These results suggest that pedunculated spines and stubby spines might play complementary roles in synaptic integration. Finally, we analyze how varicosities affect voltage diffusion in dendrites and discuss their impact on the spatiotemporal integration of synaptic input.

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87. Spike Train Probability Models for Stimulus-Driven Leaky Integrate-and-Fire Neurons

Shinsuke Koyama, Robert E. Kass

https://doi.org/10.1162/neco.2008.06-07-540 | pp. 1776-1795

Mathematical models of neurons are widely used to improve understanding of neuronal spiking behavior. These models can produce artificial spike trains that resemble actual spike train data in important ways, but they are not very easy to apply to the analysis of spike train data. Instead, statistical methods based on point process models of spike trains provide a wide range of data-analytical techniques. Two simplified point process models have been introduced in the literature: the time-rescaled renewal process (TRRP) and the multiplicative inhomogeneous Markov interval (m-IMI) model. In this letter we investigate the extent to which the TRRP and m-IMI models are able to fit spike trains produced by stimulus-driven leaky integrate-and-fire (LIF) neurons. With a constant stimulus, the LIF spike train is a renewal process, and the m-IMI and TRRP

models will describe accurately the LIF spike train variability. With a time-varying stimulus, the probability of spiking under all three of these models depends on both the experimental clock time relative to the stimulus and the time since the previous spike, but it does so differently for the LIF, m-IMI, and TRRP models. We assessed the distance between the LIF model and each of the two empirical models in the presence of a time-varying stimulus. We found that while lack of fit of a Poisson model to LIF spike train data can be evident even in small samples, the m-IMI and TRRP models tend to fit well, and much larger samples are required before there is statistical evidence of lack of fit of the m-IMI or TRRP models. We also found that when the mean of the stimulus varies across time, the m-IMI model provides a better fit to the LIF data than the TRRP, and when the variance of the stimulus varies across time, the TRRP provides the better fit.

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88. Parameter Estimation for α-GMM Based on Maximum Likelihood Criterion

Dalei Wu

https://doi.org/10.1162/neco.2008.04-08-776 | pp. 1776-1795

Alpha-integration and alpha-GMM have been recently proposed for integrated stochastic modeling. However, there has not been an approach to date for estimating model parameters for alpha-GMM in a statistical way, based on a set of training data. In this letter, parameter updating formulas are mathematically derived based on maximum likelihood criterion using an adapted expectation-maximization algorithm. With this method, model parameters for alpha-GMM are reestimated in an iterative way. The updating formulas were found to be simple and systematically compatible with the GMM equations. This advantage renders the alpha-GMM a superset of the GMM but with similar computational complexity. This method has been effectively applied to realistic speaker recognition applications.

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89. Temporal Coding: Assembly Formation Through Constructive Interference

Thomas Burwick

https://doi.org/10.1162/neco.2008.09-06-342 | pp. 1796-1820

Temporal coding is studied for an oscillatory neural network model with synchronization and acceleration. The latter mechanism refers to increasing (decreasing) the phase velocity of each unit for stronger (weaker) or more coherent (decoherent) input from the other units. It has been demonstrated that acceleration generates the desynchronization that is needed for self-organized segmentation of two overlapping patterns. In this letter, we continue the discussion of this remarkable feature, giving also an example with several overlapping patterns. Due to acceleration, Hebbian memory implies a frequency spectrum for pure pattern states, defined as coherent patterns with decoherent overlapping patterns. With reference to this frequency spectrum and related frequency bands, the process of pattern retrieval, corresponding to the formation of temporal coding assemblies, is described as resulting from constructive interference (with frequency differences due to acceleration) and phase locking (due to synchronization).

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90. Fast Recursive Filters for Simulating Nonlinear Dynamic Systems

J. H. van Hateren

https://doi.org/10.1162/neco.2008.04-07-506 | pp. 1821-1846

A fast and accurate computational scheme for simulating nonlinear dynamic systems is presented. The scheme assumes that the system can be represented by a combination of components of only two different types: first-order low-pass filters and static nonlinearities. The parameters of these filters and nonlinearities may depend on system variables, and the topology of the system may be complex, including feedback. Several examples taken from neuroscience are given: phototransduction, photopigment bleaching, and spike generation according to the Hodgkin-Huxley equations. The scheme uses two slightly different forms of autoregressive filters, with an implicit delay of zero for feedforward control and an implicit delay of half a sample distance for feedback control. On a fairly complex model of the macaque retinal horizontal cell, it computes, for a given level of accuracy, one to two orders of magnitude faster than the fourth-order Runge-Kutta. The computational scheme has minimal memory requirements and is also suited for computation on a stream processor, such as a graphical processing unit.

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91. Adaptive Integration in the Visual Cortex by Depressing Recurrent Cortical Circuits

Mark C. W. van Rossum, Matthijs A. A. van der Meer, Dengke Xiao, Mike W. Oram

https://doi.org/10.1162/neco.2008.06-07-546 | pp. 1847-1872

Neurons in the visual cortex receive a large amount of input from recurrent connections, yet the functional role of these connections remains unclear. Here we explore networks with strong recurrence in a computational model and show that short-term depression of the synapses in the recurrent loops implements an adaptive filter. This allows the visual system to respond reliably to deteriorated stimuli yet quickly to high-quality stimuli. For low-contrast stimuli, the model predicts long response latencies, whereas latencies are short for high-contrast stimuli. This is consistent with physiological data showing that in higher visual areas, latencies can increase more than 100 ms at low contrast compared to high contrast. Moreover, when presented with briefly flashed stimuli, the model predicts stereotypical responses that outlast the stimulus, again consistent with physiological findings. The adaptive properties of the model suggest that the abundant recurrent connections found in visual cortex serve to adapt the network's time constant in accordance with the stimulus and normalizes neuronal signals such that processing is as fast as possible while maintaining reliability.

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92. Spectral Algorithms for Supervised Learning

L. Lo Gerfo, Lorenzo Rosasco, Francesca Odone, Ernesto De Vito, Alessandro Verri

https://doi.org/10.1162/neco.2008.05-07-517 | pp. 1873-1897

We discuss how a large class of regularization methods, collectively known as spectral regularization and originally designed for solving ill-posed inverse problems, gives rise to regularized learning algorithms. All of these algorithms are consistent kernel methods that can be easily implemented. The intuition behind their derivation is that the same principle allowing for the numerical stabilization of a matrix inversion problem is crucial to avoid overfitting. The various methods have a common derivation but different computational and theoretical properties. We describe examples of such algorithms, analyze their classification performance on several data sets and discuss their applicability to real-world problems.

93. 3D Axon Structure Extraction and Analysis in Confocal Fluorescence Microscopy Images

Yong Zhang, Xiaobo Zhou, Ju Lu, Jeff W. Lichtman, Donald Adjeroh, Stephen T.C. Wong

https://doi.org/10.1162/neco.2008.05-07-519 | pp. 1899-1927

The morphological properties of axons, such as their branching patterns and oriented structures, are of great interest for biologists in the study of the synaptic connectivity of neurons. In these studies, researchers use triple immunofluorescent confocal microscopy to record morphological changes of neuronal processes. Three-dimensional (3D) microscopy image analysis is then required to extract morphological features of the neuronal structures. In this article, we propose a highly automated 3D centerline extraction tool to assist in this task. For this project, the most difficult part is that some axons are overlapping such that the boundaries distinguishing them are barely visible. Our approach combines a 3D dynamic programming (DP) technique and marker-controlled watershed algorithm to solve this problem. The approach consists of tracking and updating along the navigation directions of multiple axons simultaneously. The experimental results show that the proposed method can rapidly and accurately extract multiple axon centerlines and can handle complicated axon structures such as cross-over sections and overlapping objects.

94. Optimizing One-Shot Learning with Binary Synapses

Sandro Romani, Daniel J. Amit, Yali Amit

https://doi.org/10.1162/neco.2008.10-07-618 | pp. 1928-1950

A network of excitatory synapses trained with a conservative version of Hebbian learning is used as a model for recognizing the familiarity of thousands of once-seen stimuli from those never seen before. Such networks were initially proposed for modeling memory retrieval (selective delay activity). We show that the same framework allows the incorporation of both familiarity recognition and memory retrieval, and estimate the network's capacity. In the case of binary neurons, we extend the analysis of Amit and Fusi (1994) to obtain capacity limits based on computations of signal-to-noise ratio of the field difference between selective and non-selective neurons of learned signals. We show that with fast learning (potentiation probability approximately 1), the most recently learned patterns can be retrieved in working memory (selective delay activity). A much higher number of once-seen learned patterns elicit a realistic familiarity signal in the presence of an external field. With potentiation probability much less than 1 (slow learning), memory retrieval disappears, whereas familiarity recognition capacity is maintained at a similarly high level. This analysis is corroborated in simulations. For analog neurons, where such analysis is more difficult, we simplify the capacity analysis by studying the excess number of potentiated synapses above the steady-state distribution. In this framework, we derive the optimal constraint between potentiation and depression probabilities that maximizes the capacity.

95. Stochastic Synchrony of Chaos in a Pulse-Coupled Neural Network with Both Chemical and Electrical Synapses Among Inhibitory Neurons

Takashi Kanamaru, Kazuyuki Aihara

https://doi.org/10.1162/neco.2008.05-07-516 | pp. 1951-1972

The synchronous firing of neurons in a pulse-coupled neural network composed of excitatory and inhibitory neurons is analyzed. The neurons are connected by both chemical synapses and electrical synapses among the inhibitory neurons. When electrical synapses are introduced, periodically synchronized firing as well as chaotically synchronized firing is widely observed. Moreover, we find stochastic synchrony where the ensemble-averaged dynamics shows synchronization in the network but each neuron has a low firing rate and the firing of the neurons seems to be stochastic. Stochastic synchrony of chaos corresponding to a chaotic attractor is also found.

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96. Can Spike Coordination Be Differentiated from Rate Covariation?

Benjamin Staude, Stefan Rotter, Sonja Grün

https://doi.org/10.1162/neco.2008.06-07-550 | pp. 1973-1999

There has been a long and lively debate on whether rate covariance and temporal coordination of spikes, regarded as potential origins for correlations in cortical spike signals, fulfill different roles in the cortical code. In this context, studies that report spike coordination have often been criticized for ignoring fast nonstationarities, which would result in wrongly assigned spike coordination. The underlying hypothesis of this critique is that spike coordination is essentially identical to rate covariation, only on a shorter timescale. This study investigates the validity of this critique. We provide a decomposition for the cross-correlation function of doubly stochastic point processes, where each of the components corresponds precisely to the concepts of dependence under investigation. This allows us to correct the correlation function for rate effects, which implies that spike coordination and rate covariation are statistically separable concepts of dependence. Furthermore, we present direct and intuitive model implementations of the discussed concepts and illustrate that their difference is not a matter of timescale. Analysis of data generated by our models and analytical description of the relevant estimators reveals, however, that spike coordination dramatically influences the accuracy of rate covariance estimation. As a consequence, extreme parameter combinations can lead to situations where the concept of dependence cannot be identified empirically. However, for a wide range of parameters, the concept of dependence underlying a given data set can be identified regardless of its timescale.

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97. Recoding Patterns of Sensory Input: Higher-Order Features and the Function of Nonlinear Dendritic Trees

Paul Rhodes

https://doi.org/10.1162/neco.2008.04-07-511 | pp. 2000-2036

Here analytical and simulation results are presented characterizing the recoding arising when overlapping patterns of sensor input impinge on an array of model neurons with branched thresholded dendritic trees. Thus, the neural units employed are intended to capture the integrative behavior of pyramidal cells that sustain isolated Na(+) or NMDA spikes in their branches. Given a defined set of sensor vectors, equations were derived for the probability of firing of both branches and neurons and for the expected overlap between the neural firing patterns triggered by two afferent patterns of given overlap. Thus, both the sparseness of the neural representation and the orthogonalization of overlapping vectors were computed. Simulations were then performed with an array of 1000 neurons comprising 30,000 branches to verify the analytical results and confirm their applicability to systems (which include any practicable artificial system) in which the

combinatorically possible branches and neurons are severely subsampled. A means of readout and a measure of discrimination performance were provided so that the accuracy of discrimination among overlapping sensor vectors could be optimized as a function of neuron structure parameters. Good performance required both orthogonalization of the afferent patterns, so that discrimination was accurate and free of interference, and maintenance of a minimum level of neural activity, so that some neurons fired in response to each sensor pattern. It is shown that the discrimination performance achieved by arrays of neurons with branched dendritic trees could not be reached with single-compartment units, regardless of how many of the latter are used. The analytical results furnish a benchmark against which to measure further enhancements in the performance of subsequent simulated systems incorporating local neural mechanisms which, while often less amenable to closed-form analysis, are ubiquitous in biological neural circuitry.

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98. Oscillations and Spiking Pairs: Behavior of a Neuronal Model with STDP Learning

Xi Shen, Xiaobin Lin, Philippe De Wilde

https://doi.org/10.1162/neco.2008.08-06-317 | pp. 2037-2069

In a biologically plausible but computationally simplified integrate-and-fire neuronal population, it is observed that transient synchronized spikes can occur repeatedly. However, groups with different properties exhibit different periods and different patterns of synchrony. We include learning mechanisms in these models. The effects of spike-timing-dependent plasticity have been known to play a distinct role in information processing in the central nervous system for several years. In this letter, neuronal models with dynamical synapses are constructed, and we analyze the effect of STDP on collective network behavior, such as oscillatory activity, weight distribution, and spike timing precision. We comment on how information is encoded by the neuronal signaling, when synchrony groups may appear, and what could contribute to the uncertainty in decision making.

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99. Dynamic Moment Analysis of the Extracellular Electric Field of a Biologically Realistic Spiking Neuron

Joshua N. Milstein, Christof Koch

https://doi.org/10.1162/neco.2008.06-07-537 | pp. 2070-2084

Based on the membrane currents generated by an action potential in a biologically realistic model of a pyramidal, hippocampal cell within rat CA1, we perform a moment expansion of the extracellular field potential. We decompose the potential into both inverse and classical moments and show that this method is a rapid and efficient way to calculate the extracellular field both near and far from the cell body. The action potential gives rise to a large quadrupole moment that contributes to the extracellular field up to distances of almost 1 cm. This method will serve as a starting point in connecting the microscopic generation of electric fields at the level of neurons to macroscopic observables such as the local field potential.

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100. Sensitivity Derivatives for Flexible Sensorimotor Learning

Mohamed Abdelghani, Timothy Lillicrap, Douglas Tweed
https://doi.org/10.1162/neco.2008.04-07-507 | pp. 2085-2111

To learn effectively, an adaptive controller needs to know its sensitivity derivatives--the variables that quantify how system performance depends on the commands from the controller. In the case of biological sensorimotor control, no one has explained how those derivatives themselves might be learned, and some authors suggest they are not learned at all but are known innately. Here we show that this knowledge cannot be solely innate, given the adaptive flexibility of neural systems. And we show how it could be learned using forms of information transport that are available in the brain. The mechanism, which we call implicit supervision, helps explain the flexibility and speed of sensorimotor learning and our ability to cope with high-dimensional work spaces and tools.

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101. Algorithms for Sparse Nonnegative Tucker Decompositions

Morten Mørup, Lars Kai Hansen, Sidse Arnfred
https://doi.org/10.1162/neco.2008.11-06-407 | pp. 2112-2131

There is a increasing interest in analysis of large-scale multiway data. The concept of multiway data refers to arrays of data with more than two dimensions, that is, taking the form of tensors. To analyze such data, decomposition techniques are widely used. The two most common decompositions for tensors are the Tucker model and the more restricted PARAFAC model. Both models can be viewed as generalizations of the regular factor analysis to data of more than two modalities. Nonnegative matrix factorization (NMF), in conjunction with sparse coding, has recently been given much attention due to its part-based and easy interpretable representation. While NMF has been extended to the PARAFAC model, no such attempt has been done to extend NMF to the Tucker model. However, if the tensor data analyzed are nonnegative, it may well be relevant to consider purely additive (i.e., nonnegative) Tucker decompositions). To reduce ambiguities of this type of decomposition, we develop updates that can impose sparseness in any combination of modalities, hence, proposed algorithms for sparse nonnegative Tucker decompositions (SN-TUCKER). We demonstrate how the proposed algorithms are superior to existing algorithms for Tucker decompositions when the data and interactions can be considered nonnegative. We further illustrate how sparse coding can help identify what model (PARAFAC or Tucker) is more appropriate for the data as well as to select the number of components by turning off excess components. The algorithms for SN-TUCKER can be downloaded from Mørup (2007).

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102. Dependence of Neuronal Correlations on Filter Characteristics and Marginal Spike Train Statistics

Tom Tetzlaff, Stefan Rotter, Eran Stark, Moshe Abeles, Ad Aertsen, Markus Diesmann
https://doi.org/10.1162/neco.2008.05-07-525 | pp. 2133-2184

Correlated neural activity has been observed at various signal levels (e.g., spike count, membrane potential, local field potential, EEG, fMRI BOLD). Most of these signals can be considered as superpositions of spike trains filtered by components of the neural system (synapses, membranes) and the measurement process. It is largely unknown how the spike train correlation structure is altered by this filtering and what the consequences for the dynamics of the system and for the interpretation of measured correlations are. In this

study, we focus on linearly filtered spike trains and particularly consider correlations caused by overlapping presynaptic neuron populations. We demonstrate that correlation functions and statistical second-order measures like the variance, the covariance, and the correlation coefficient generally exhibit a complex dependence on the filter properties and the statistics of the presynaptic spike trains. We point out that both contributions can play a significant role in modulating the interaction strength between neurons or neuron populations. In many applications, the coherence allows a filter-independent quantification of correlated activity. In different network models, we discuss the estimation of network connectivity from the high-frequency coherence of simultaneous intracellular recordings of pairs of neurons.

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103. Correlations and Population Dynamics in Cortical Networks

Birgit Kriener, Tom Tetzlaff, Ad Aertsen, Markus Diesmann, Stefan Rotter

https://doi.org/10.1162/neco.2008.02-07-474 | pp. 2185-2226

The function of cortical networks depends on the collective interplay between neurons and neuronal populations, which is reflected in the correlation of signals that can be recorded at different levels. To correctly interpret these observations it is important to understand the origin of neuronal correlations. Here we study how cells in large recurrent networks of excitatory and inhibitory neurons interact and how the associated correlations affect stationary states of idle network activity. We demonstrate that the structure of the connectivity matrix of such networks induces considerable correlations between synaptic currents as well as between subthreshold membrane potentials, provided Dale's principle is respected. If, in contrast, synaptic weights are randomly distributed, input correlations can vanish, even for densely connected networks. Although correlations are strongly attenuated when proceeding from membrane potentials to action potentials (spikes), the resulting weak correlations in the spike output can cause substantial fluctuations in the population activity, even in highly diluted networks. We show that simple mean-field models that take the structure of the coupling matrix into account can adequately describe the power spectra of the population activity. The consequences of Dale's principle on correlations and rate fluctuations are discussed in the light of recent experimental findings.

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104. On Exponential Convergence Conditions of an Extended Projection Neural Network

Youshen Xia, Dongyi Ye

https://doi.org/10.1162/neco.2008.10-06-389 | pp. 2227-2237

Recently the extended projection neural network was proposed to solve constrained monotone variational inequality problems and a class of constrained nonmonotontic variational inequality problems. Its exponential convergence was developed under the positive definiteness condition of the Jacobian matrix of the nonlinear mapping. This note proposes new results on the exponential convergence of the output trajectory of the extended projection neural network under the weak conditions that the Jacobian matrix of the nonlinear mapping may be positive semidefinite or not. Therefore, new results further demonstrate that the extended projection neural network has a fast convergence rate when solving a class of constrained monotone variational inequality problems and nonmonotonic variational inequality problems. Illustrative examples show the significance of the obtained results.

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105. Contrastive Divergence in Gaussian Diffusions

Javier R. Movellan

https://doi.org/10.1162/neco.2008.01-07-430 | pp. 2238-2252

This letter presents an analysis of the contrastive divergence (CD) learning algorithm when applied to continuous-time linear stochastic neural networks. For this case, powerful techniques exist that allow a detailed analysis of the behavior of CD. The analysis shows that CD converges to maximum likelihood solutions only when the network structure is such that it can match the first moments of the desired distribution. Otherwise, CD can converge to solutions arbitrarily different from the log-likelihood solutions, or they can even diverge. This result suggests the need to improve our theoretical understanding of the conditions under which CD is expected to be well behaved and the conditions under which it may fail. In, addition the results point to practical ideas on how to improve the performance of CD.

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106. Temporal Dynamics of Rate-Based Synaptic Plasticity Rules in a Stochastic Model of Spike-Timing-Dependent Plasticity

Terry Elliott

https://doi.org/10.1162/neco.2008.06-07-555 | pp. 2253-2307

In a recently proposed, stochastic model of spike-timing-dependent plasticity, we derived general expressions for the expected change in synaptic strength, DeltaS n, induced by a typical sequence of precisely n spikes. We found that the rules DeltaS n, n >or= 3, exhibit regions of parameter space in which stable, competitive interactions between afferents are present, leading to the activity-dependent segregation of afferents on their targets. The rules DeltaS n, however, allow an indefinite period of time to elapse for the occurrence of precisely n spikes, while most measurements of changes in synaptic strength are conducted over definite periods of time during which a potentially unknown number of spikes may occur. Here, therefore, we derive an expression, DeltaS(t), for the expected change in synaptic strength of a synapse experiencing an average sequence of spikes of typical length occurring during a fixed period of time, t. We find that the resulting synaptic plasticity rule Delta S(t) exhibits a number of remarkable properties. It is an entirely self-stabilizing learning rule in all regions of parameter space. Further, its parameter space is carved up into three distinct, contiguous regions in which the exhibited synaptic interactions undergo different transitions as the time t is increased. In one region, the synaptic dynamics change from noncompetitive to competitive to entirely depressing. In a second region, the dynamics change from noncompetitive to competitive without the second transition to entirely depressing dynamics. In a third region, the dynamics are always noncompetitive. The locations of these regions are not fixed in parameter space but may be modified by changing the mean presynaptic firing rates. Thus, neurons may be moved among these three different regions and so exhibit different sets of synaptic dynamics depending on their mean firing rates.

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107. Random Neural Networks with Synchronized Interactions

Erol Gelenbe, Stelios Timotheou

https://doi.org/10.1162/neco.2008.04-07-509 | pp. 2308-2324

Large-scale distributed systems, such as natural neuronal and artificial systems, have many local interconnections, but they often also have the ability to propagate information very fast over relatively large

distances. Mechanisms that enable such behavior include very long physical signaling paths and possibly saccades of synchronous behavior that may propagate across a network. This letter studies the modeling of such behaviors in neuronal networks and develops a related learning algorithm. This is done in the context of the random neural network (RNN), a probabilistic model with a well-developed mathematical theory, which was inspired by the apparently stochastic spiking behavior of certain natural neuronal systems. Thus, we develop an extension of the RNN to the case when synchronous interactions can occur, leading to synchronous firing by large ensembles of cells. We also present an O(N3) gradient descent learning algorithm for an N-cell recurrent network having both conventional excitatory-inhibitory interactions and synchronous interactions. Finally, the model and its learning algorithm are applied to a resource allocation problem that is NP-hard and requires fast approximate decisions.

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108. Encoding and Decoding Spikes for Dynamic Stimuli

Rama Natarajan, Quentin J. M. Huys, Peter Dayan, Richard S. Zemel

https://doi.org/10.1162/neco.2008.01-07-436 | pp. 2325-2360

Naturally occurring sensory stimuli are dynamic. In this letter, we consider how spiking neural populations might transmit information about continuous dynamic stimulus variables. The combination of simple encoders and temporal stimulus correlations leads to a code in which information is not readily available to downstream neurons. Here, we explore a complex encoder that is paired with a simple decoder that allows representation and manipulation of the dynamic information in neural systems. The encoder we present takes the form of a biologically plausible recurrent spiking neural network where the output population recodes its inputs to produce spikes that are independently decodeable. We show that this network can be learned in a supervised manner by a simple local learning rule.

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109. A (Somewhat) New Solution to the Variable Binding Problem

Leon Barrett, Jerome A. Feldman, Liam Mac Dermed

https://doi.org/10.1162/neco.2008.07-06-290 | pp. 2361-2378

To perform automatic, unconscious inference, the human brain must solve the binding problem by correctly grouping properties with objects. Temporal binding models like SHRUTI already suggest much of how this might be done in a connectionist and localist way by using temporal synchrony. We propose a set of alternatives to temporal synchrony mechanisms that instead use short signatures. This serves two functions: it allows us to explore an additional biologically plausible alternative, and it allows us to extend and improve the capabilities of these models. These extensions model the human ability to both perform unification and handle multiple instantiations of logical terms. To verify our model's feasibility, we simulate it with a computer system modeling simple, neuron-like computations.

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110. Neural Integrator: A Sandpile Model

Maxim Nikitchenko, Alexei A. Koulakov

https://doi.org/10.1162/neco.2008.12-06-416 | pp. 2379-2417

We investigated a model for the neural integrator based on hysteretic units connected by positive feedback. Hysteresis is assumed to emerge from the intrinsic properties of the cells. We consider the recurrent networks containing either bistable or multistable neurons. We apply our analysis to the oculomotor velocity-to-position neural integrator that calculates eye positions using the inputs that carry information about eye angular velocity. By analyzing this system in the parameter space, we show the following. The direction of hysteresis in the neuronal response may be reversed for the system with recurrent connections compared to the case of unconnected neurons. Thus, for the NMDA receptor-based bistability, the firing rates after ON saccades may be higher than after OFF saccades for the same eye position. The reversal of hysteresis occurs in this model only when the size of hysteresis differs from neuron to neuron. We also relate the macroscopic leak time constant of the integrator to the rate of microscopic spontaneous noise-driven transitions in the hysteretic units. Finally, we investigate the conditions under which the hysteretic integrator may have no threshold for integration.

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111. Type I and Type II Neuron Models Are Selectively Driven by Differential Stimulus Features

Germán Mato, Inés Samengo

https://doi.org/10.1162/neco.2008.10-07-632 | pp. 2418-2440

Neurons in the nervous system exhibit an outstanding variety of morphological and physiological properties. However, close to threshold, this remarkable richness may be grouped succinctly into two basic types of excitability, often referred to as type I and type II. The dynamical traits of these two neuron types have been extensively characterized. It would be interesting, however, to understand the information-processing consequences of their dynamical properties. To that end, here we determine the differences between the stimulus features inducing firing in type I and type II neurons. We work with both realistic conductance-based models and minimal normal forms. We conclude that type I neurons fire in response to scale-free depolarizing stimuli. Type II neurons, instead, are most efficiently driven by input stimuli containing both depolarizing and hyperpolarizing phases, with significant power in the frequency band corresponding to the intrinsic frequencies of the cell.

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112. Rapid Convergence to Feature Layer Correspondences

Jörg Lücke, Christian Keck, Christoph von der Malsburg

https://doi.org/10.1162/neco.2008.06-07-539 | pp. 2441-2463

We describe a neural network able to rapidly establish correspondence between neural feature layers. Each of the network's two layers consists of interconnected cortical columns, and each column consists of inhibitorily coupled subpopulations of excitatory neurons. The dynamics of the system builds on a dynamic model of a single column, which is consistent with recent experimental findings. The network realizes dynamic links between its layers with the help of specialized columns that evaluate similarities between the activity distributions of local feature cell populations, are subject to a topology constraint, and can gate the transfer of feature information between the neural layers. The system can robustly be applied to natural images, and correspondences are found in time intervals estimated to be smaller than 100 ms in physiological terms.

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113. Normalization Enables Robust Validation of Disparity Estimates from Neural Populations

E.K.C. Tsang, Bertram E. Shi

https://doi.org/10.1162/neco.2008.05-07-532 | pp. 2464-2490

Binocular fusion takes place over a limited region smaller than one degree of visual angle (Panum's fusional area), which is on the order of the range of preferred disparities measured in populations of disparity-tuned neurons in the visual cortex. However, the actual range of binocular disparities encountered in natural scenes extends over tens of degrees. This discrepancy suggests that there must be a mechanism for detecting whether the stimulus disparity is inside or outside the range of the preferred disparities in the population. Here, we compare the efficacy of several features derived from the population responses of phase-tuned disparity energy neurons in differentiating between in-range and out-of-range disparities. Interestingly, some features that might be appealing at first glance, such as the average activation across the population and the difference between the peak and average responses, actually perform poorly. On the other hand, normalizing the difference between the peak and average responses results in a reliable indicator. Using a probabilistic model of the population responses, we improve classification accuracy by combining multiple features. A decision rule that combines the normalized peak to average difference and the peak location significantly improves performance over decision rules based on either measure in isolation. In addition, classifiers using normalized difference are also robust to mismatch between the image statistics assumed by the model and the actual image statistics.

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114. Understanding Parkinsonian Handwriting Through a Computational Model of Basal Ganglia

Gangadhar Garipelli, Denny Joseph, V. Srinivasa Chakravarthy

https://doi.org/10.1162/neco.2008.03-07-498 | pp. 2491-2525

Handwriting in Parkinson's disease (PD) is typically characterized by micrographia, jagged line contour, and unusual fluctuations in pen tip velocity. Although PD handwriting features have been used for diagnostics, they are not based on a signaling model of basal ganglia (BG). In this letter, we present a computational model of handwriting generation that highlights the role of BG. When PD conditions like reduced dopamine and altered dynamics of the subthalamic nucleus and globus pallidus externa subsystems are simulated, the handwriting produced by the model manifested characteristic PD handwriting distortions like micrographia and velocity fluctuations. Our approach to PD modeling is in tune with the perspective that PD is a dynamic disease.

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115. Sparse Coding via Thresholding and Local Competition in Neural Circuits

Christopher J. Rozell, Don H. Johnson, Richard G. Baraniuk, Bruno A. Olshausen

https://doi.org/10.1162/neco.2008.03-07-486 | pp. 2526-2563

While evidence indicates that neural systems may be employing sparse approximations to represent sensed stimuli, the mechanisms underlying this ability are not understood. We describe a locally competitive algorithm (LCA) that solves a collection of sparse coding principles minimizing a weighted combination of

mean-squared error and a coefficient cost function. LCAs are designed to be implemented in a dynamical system composed of many neuron-like elements operating in parallel. These algorithms use thresholding functions to induce local (usually one-way) inhibitory competitions between nodes to produce sparse representations. LCAs produce coefficients with sparsity levels comparable to the most popular centralized sparse coding algorithms while being readily suited for neural implementation. Additionally, LCA coefficients for video sequences demonstrate inertial properties that are both qualitatively and quantitatively more regular (i.e., smoother and more predictable) than the coefficients produced by greedy algorithms.

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116. Adaptive Classification of Temporal Signals in Fixed-Weight Recurrent Neural Networks: An Existence Proof

Ivan Tyukin, Danil Prokhorov, Cees van Leeuwen

https://doi.org/10.1162/neco.2008.05-07-527 | pp. 2564-2596

Recurrent neural networks with fixed weights have been shown in practice to successfully classify adaptively signals that vary as a function of time in the presence of additive noise and parametric perturbations. We address the question: Can this ability be explained theoretically? We provide a mathematical proof that these networks have this ability even when parametric perturbations enter the signals nonlinearly. The restrictions that we impose on the signals to be classified are that they satisfy an assumption of nondegeneracy and that noise amplitude is sufficiently small. Further, we demonstrate that the recurrent neural networks may not only classify uncertain signals adaptively but also can recover the values of uncertain parameters of the signals, up to their equivalence classes.

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117. Latent Features in Similarity Judgments: A Nonparametric Bayesian Approach

Danielle Navarro, Thomas L. Griffiths

https://doi.org/10.1162/neco.2008.04-07-504 | pp. 2597-2628

One of the central problems in cognitive science is determining the mental representations that underlie human inferences. Solutions to this problem often rely on the analysis of subjective similarity judgments, on the assumption that recognizing likenesses between people, objects, and events is crucial to everyday inference. One such solution is provided by the additive clustering model, which is widely used to infer the features of a set of stimuli from their similarities, on the assumption that similarity is a weighted linear function of common features. Existing approaches for implementing additive clustering often lack a complete framework for statistical inference, particularly with respect to choosing the number of features. To address these problems, this article develops a fully Bayesian formulation of the additive clustering model, using methods from nonparametric Bayesian statistics to allow the number of features to vary. We use this to explore several approaches to parameter estimation, showing that the nonparametric Bayesian approach provides a straightforward way to obtain estimates of both the number of features and their importance.

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118. Deep, Narrow Sigmoid Belief Networks Are Universal Approximators

Ilya Sutskever, Geoffrey E. Hinton

https://doi.org/10.1162/neco.2008.12-07-661 | pp. 2629-2636

In this note, we show that exponentially deep belief networks can approximate any distribution over binary vectors to arbitrary accuracy, even when the width of each layer is limited to the dimensionality of the data. We further show that such networks can be greedily learned in an easy yet impractical way.

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119. Exact Solutions for Rate and Synchrony in Recurrent Networks of Coincidence Detectors

Shawn Mikula, Ernst Niebur

https://doi.org/10.1162/neco.2008.07-07-570 | pp. 2637-2661

We provide analytical solutions for mean firing rates and cross-correlations of coincidence detector neurons in recurrent networks with excitatory or inhibitory connectivity, with rate-modulated steady-state spiking inputs. We use discrete-time finite-state Markov chains to represent network state transition probabilities, which are subsequently used to derive exact analytical solutions for mean firing rates and cross-correlations. As illustrated in several examples, the method can be used for modeling cortical microcircuits and clarifying single-neuron and population coding mechanisms. We also demonstrate that increasing firing rates do not necessarily translate into increasing cross-correlations, though our results do support the contention that firing rates and cross-correlations are likely to be coupled. Our analytical solutions underscore the complexity of the relationship between firing rates and cross-correlations.

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120. Mutual Information Expansion for Studying the Role of Correlations in Population Codes: How Important Are Autocorrelations?

Alessandro Scaglione, Guglielmo Foffani, G. Scannella, S. Cerutti, Karen A. Moxon

https://doi.org/10.1162/neco.2008.08-07-595 | pp. 2662-2695

The role of correlations in the activity of neural populations responding to a set of stimuli can be studied within an information theory framework. Regardless of whether one approaches the problem from an encoding or decoding perspective, the main measures used to study the role of correlations can be derived from a common source: the expansion of the mutual information. Two main formalisms of mutual information expansion have been proposed: the series expansion and the exact breakdown. Here we clarify that these two formalisms have a different representation of autocorrelations, so that even when the total information estimated differs by less than 1%, individual terms can diverge. More precisely, the series expansion explicitly evaluates the informational contribution of autocorrelations in the count of spikes, that is, count autocorrelations, whereas the exact breakdown does not. We propose a new formalism of mutual information expansion, the Poisson exact breakdown, which introduces Poisson equivalents in order to explicitly evaluate the informational contribution of count autocorrelations with no approximation involved. Because several widely employed manipulations of spike trains, most notably binning and pooling, alter the structure of count autocorrelations, the new formalism can provide a useful general framework for studying the role of correlations in population codes.

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121. Parameters of the Diffusion Leaky Integrate-and-Fire Neuronal Model for a Slowly Fluctuating Signal

Umberto Picchini, Susanne Ditlevsen, Andrea De Gaetano, Petr Lánský

https://doi.org/10.1162/neco.2008.11-07-653 | pp. 2696-2714

Stochastic leaky integrate-and-fire (LIF) neuronal models are common theoretical tools for studying properties of real neuronal systems. Experimental data of frequently sampled membrane potential measurements between spikes show that the assumption of constant parameter values is not realistic and that some (random) fluctuations are occurring. In this letter, we extend the stochastic LIF model, allowing a noise source determining slow fluctuations in the signal. This is achieved by adding a random variable to one of the parameters characterizing the neuronal input, considering each interspike interval (ISI) as an independent experimental unit with a different realization of this random variable. In this way, the variation of the neuronal input is split into fast (within-interval) and slow (between-intervals) components. A parameter estimation method is proposed, allowing the parameters to be estimated simultaneously over the entire data set. This increases the statistical power, and the average estimate over all ISIs will be improved in the sense of decreased variance of the estimator compared to previous approaches, where the estimation has been conducted on each individual ISI. The results obtained on real data show good agreement with classical regression methods.

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122. Faithful Representation of Stimuli with a Population of Integrate-and-Fire Neurons

Aurel A. Lazar, Eftychios A. Pnevmatikakis

https://doi.org/10.1162/neco.2008.06-07-559 | pp. 2715-2744

We consider a formal model of stimulus encoding with a circuit consisting of a bank of filters and an ensemble of integrate-and-fire neurons. Such models arise in olfactory systems, vision, and hearing. We demonstrate that bandlimited stimuli can be faithfully represented with spike trains generated by the ensemble of neurons. We provide a stimulus reconstruction scheme based on the spike times of the ensemble of neurons and derive conditions for perfect recovery. The key result calls for the spike density of the neural population to be above the Nyquist rate. We also show that recovery is perfect if the number of neurons in the population is larger than a threshold value. Increasing the number of neurons to achieve a faithful representation of the sensory world is consistent with basic neurobiological thought. Finally we demonstrate that in general, the problem of faithful recovery of stimuli from the spike train of single neurons is ill posed. The stimulus can be recovered, however, from the information contained in the spike train of a population of neurons.

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123. Just-in-Time Connectivity for Large Spiking Networks

William W. Lytton, Ahmet Omurtag, Samuel A. Neymotin, Michael L. Hines

https://doi.org/10.1162/neco.2008.10-07-622 | pp. 2745-2756

The scale of large neuronal network simulations is memory limited due to the need to store connectivity information: connectivity storage grows as the square of neuron number up to anatomically relevant limits. Using the NEURON simulator as a discrete-event simulator (no integration), we explored the consequences

of avoiding the space costs of connectivity through regenerating connectivity parameters when needed: just in time after a presynaptic cell fires. We explored various strategies for automated generation of one or more of the basic static connectivity parameters: delays, postsynaptic cell identities, and weights, as well as run-time connectivity state: the event queue. Comparison of the JitCon implementation to NEURON's standard NetCon connectivity method showed substantial space savings, with associated run-time penalty. Although JitCon saved space by eliminating connectivity parameters, larger simulations were still memory limited due to growth of the synaptic event queue. We therefore designed a JitEvent algorithm that added items to the queue only when required: instead of alerting multiple postsynaptic cells, a spiking presynaptic cell posted a callback event at the shortest synaptic delay time. At the time of the callback, this same presynaptic cell directly notified the first postsynaptic cell and generated another self-callback for the next delay time. The JitEvent implementation yielded substantial additional time and space savings. We conclude that just-in-time strategies are necessary for very large network simulations but that a variety of alternative strategies should be considered whose optimality will depend on the characteristics of the simulation to be run.

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124. Simultaneous Approximations of Polynomials and Derivatives and Their Applications to Neural Networks

Yoshifusa Ito

https://doi.org/10.1162/neco.2008.03-07-494 | pp. 2757-2791

We have constructed one-hidden-layer neural networks capable of approximating polynomials and their derivatives simultaneously. Generally, optimizing neural network parameters to be trained at later steps of the BP training is more difficult than optimizing those to be trained at the first step. Taking into account this fact, we suppressed the number of parameters of the former type. We measure degree of approximation in both the uniform norm on compact sets and the Lp-norm on the whole space with respect to probability measures.

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125. Boosting Method for Local Learning in Statistical Pattern Recognition

Masanori Kawakita, Shinto Eguchi

https://doi.org/10.1162/neco.2008.06-07-549 | pp. 2792-2838

We propose a local boosting method in classification problems borrowing from an idea of the local likelihood method. Our proposal, local boosting, includes a simple device for localization for computational feasibility. We proved the Bayes risk consistency of the local boosting in the framework of Probably approximately correct learning. Inspection of the proof provides a useful viewpoint for comparing ordinary boosting and local boosting with respect to the estimation error and the approximation error. Both boosting methods have the Bayes risk consistency if their approximation errors decrease to zero. Compared to ordinary boosting, local boosting may perform better by controlling the trade-off between the estimation error and the approximation error. Ordinary boosting with complicated base classifiers or other strong classification methods, including kernel machines, may have classification performance comparable to local boosting with simple base classifiers, for example, decision stumps. Local boosting, however, has an advantage with respect to interpretability. Local boosting with simple base classifiers offers a simple way to specify which features are informative and how their values contribute to a classification rule even though locally. Several

numerical studies on real data sets confirm these advantages of local boosting.

126. A Scalable Kernel-Based Semisupervised Metric Learning Algorithm with Out-of-Sample Generalization Ability

Dit■Yan Yeung, Hong Chang, Guang Dai

https://doi.org/10.1162/neco.2008.05-07-528 | pp. 2839-2861

In recent years, metric learning in the semisupervised setting has aroused a lot of research interest. One type of semisupervised metric learning utilizes supervisory information in the form of pairwise similarity or dissimilarity constraints. However, most methods proposed so far are either limited to linear metric learning or unable to scale well with the data set size. In this letter, we propose a nonlinear metric learning method based on the kernel approach. By applying low-rank approximation to the kernel matrix, our method can handle significantly larger data sets. Moreover, our low-rank approximation scheme can naturally lead to out-of-sample generalization. Experiments performed on both artificial and real-world data show very promising results.

127. Optimization of Decision Making in Multilayer Networks: The Role of Locus Coeruleus

Eric Shea■Brown, Mark S. Gilzenrat, Jonathan D. Cohen

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Previous theoretical work has shown that a single-layer neural network can implement the optimal decision process for simple, two-alternative forced-choice (2AFC) tasks. However, it is likely that the mammalian brain comprises multilayer networks, raising the question of whether and how optimal performance can be approximated in such an architecture. Here, we present theoretical work suggesting that the noradrenergic nucleus locus coeruleus (LC) may help optimize 2AFC decision making in the brain. This is based on the observations that neurons of the LC selectively fire following the presentation of salient stimuli in decision tasks and that the corresponding release of norepinephrine can transiently increase the responsivity, or gain, of cortical processing units. We describe computational simulations that investigate the role of such gain changes in optimizing performance of 2AFC decision making. In the tasks we model, no prior cueing or knowledge of stimulus onset time is assumed. Performance is assessed in terms of the rate of correct responses over time (the reward rate). We first present the results of a single-layer model that accumulates (integrates) sensory input and implements the decision process as a threshold crossing. Gain transients, representing the modulatory effect of the LC, are driven by separate threshold crossings in this layer. We optimize over all free parameters to determine the maximum reward rate achievable by this model and compare it to the maximum reward rate when gain is held fixed. We find that the dynamic gain mechanism yields no improvement in reward for this single-layer model. We then examine a two-layer model, in which competing sensory accumulators in the first layer (capable of implementing the task relevant decision) pass activity to response accumulators in a second layer. Again, we compare a version in which threshold crossing in the first (decision) layer elicits an LC response (and a concomitant increase in gain) with a fixed-gain version of the model. Here, we find that gain transients modeling the LC phasic response yield an improvement in reward rate of 12% to 24%. Furthermore, we show that the timing characteristics of these gain transients agree with observations concerning LC firing patterns reported in recent experimental

studies. This provides converging evidence for the hypothesis that the LC optimizes processes underlying 2AFC decision making in multilayer networks.

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128. Indices for Testing Neural Codes

Jonathan D. Victor, Sheila Nirenberg

https://doi.org/10.1162/neco.2008.10-07-633 | pp. 2895-2936

One of the most critical challenges in systems neuroscience is determining the neural code. A principled framework for addressing this can be found in information theory. With this approach, one can determine whether a proposed code can account for the stimulus-response relationship. Specifically, one can compare the transmitted information between the stimulus and the hypothesized neural code with the transmitted information between the stimulus and the behavioral response. If the former is smaller than the latter (i.e., if the code cannot account for the behavior), the code can be ruled out. The information-theoretic index most widely used in this context is Shannon's mutual information. The Shannon test, however, is not ideal for this purpose: while the codes it will rule out are truly nonviable, there will be some nonviable codes that it will fail to rule out. Here we describe a wide range of alternative indices that can be used for ruling codes out. The range includes a continuum from Shannon information to measures of the performance of a Bayesian decoder. We analyze the relationship of these indices to each other and their complementary strengths and weaknesses for addressing this problem.

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129. A Mathematical Analysis of the Effects of Hebbian Learning Rules on the Dynamics and Structure of Discrete-Time Random Recurrent Neural Networks

Benoît Siri, Hugues Berry, Bruno Cessac, Bruno Delord, Mathias Quoy

https://doi.org/10.1162/neco.2008.05-07-530 | pp. 2937-2966

We present a mathematical analysis of the effects of Hebbian learning in random recurrent neural networks, with a generic Hebbian learning rule, including passive forgetting and different timescales, for neuronal activity and learning dynamics. Previous numerical work has reported that Hebbian learning drives the system from chaos to a steady state through a sequence of bifurcations. Here, we interpret these results mathematically and show that these effects, involving a complex coupling between neuronal dynamics and synaptic graph structure, can be analyzed using Jacobian matrices, which introduce both a structural and a dynamical point of view on neural network evolution. Furthermore, we show that sensitivity to a learned pattern is maximal when the largest Lyapunov exponent is close to 0. We discuss how neural networks may take advantage of this regime of high functional interest.

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130. A Fast, Streaming SIMD Extensions 2, Logistic Squashing Function

J. J. Milner, Angus Grandison

https://doi.org/10.1162/neco.2008.10-06-366 | pp. 2967-2972

Schraudolph proposed an excellent exponential approximation providing increased performance particularly suited to the logistic squashing function used within many neural networking applications. This note applies

Intel's streaming SIMD Extensions 2 (SSE2), where SIMD is single instruction multiple data, of the Pentium IV class processor to Schraudolph's technique, further increasing the performance of the logistic squashing function. It was found that the calculation of the new 32-bit SSE2 logistic squashing function described here was up to 38 times faster than the conventional exponential function and up to 16 times faster than a Schraudolph-style 32-bit method on an Intel Pentium D 3.6 GHz CPU.

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131. Gamma Oscillations in a Nonlinear Regime: A Minimal Model Approach Using Heterogeneous Integrate-and-Fire Networks

Brice Bathellier, Alan Carleton, Wulfram Gerstner

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Fast oscillations and in particular gamma-band oscillation (20-80 Hz) are commonly observed during brain function and are at the center of several neural processing theories. In many cases, mathematical analysis of fast oscillations in neural networks has been focused on the transition between irregular and oscillatory firing viewed as an instability of the asynchronous activity. But in fact, brain slice experiments as well as detailed simulations of biological neural networks have produced a large corpus of results concerning the properties of fully developed oscillations that are far from this transition point. We propose here a mathematical approach to deal with nonlinear oscillations in a network of heterogeneous or noisy integrate-and-fire neurons connected by strong inhibition. This approach involves limited mathematical complexity and gives a good sense of the oscillation mechanism, making it an interesting tool to understand fast rhythmic activity in simulated or biological neural networks. A surprising result of our approach is that under some conditions, a change of the strength of inhibition only weakly influences the period of the oscillation. This is in contrast to standard theoretical and experimental models of interneuron network gamma oscillations (ING), where frequency tightly depends on inhibition strength, but it is similar to observations made in some in vitro preparations in the hippocampus and the olfactory bulb and in some detailed network models. This result is explained by the phenomenon of suppression that is known to occur in strongly coupled oscillating inhibitory networks but had not yet been related to the behavior of oscillation frequency.

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132. Analytical and Simulation Results for the Stochastic Spatial Fitzhugh-Nagumo Model Neuron

Henry C. Tuckwell

https://doi.org/10.1162/neco.2008.08-07-585 | pp. 3003-3033

For the Fitzhugh-Nagumo system with space-time white noise, we use numerical methods to consider the generation of action potentials and the reliability of transmission in the presence of noise. The accuracy of simulated solutions is verified by comparison with known exact analytical results. Noise of small amplitude may prevent transmission directly, whereas larger-amplitude noise may also interfere by producing secondary nonlocal responses. The probability of transmission as a function of noise amplitude is found for both uniform noise and noise restricted to a patch. For certain parameter ranges, the recovery variable may be neglected to give a single-component nonlinear diffusion with space-time white noise. In this case, analytical results are obtained for small perturbations and noise, which agree well with simulation results. For the voltage variable, expressions are given for the mean, covariance, and variance and their

steady-state forms. The spectral density of the voltage is also obtained. Numerical examples are given of the difference between the properties of nonlinear and linear cables, and the validity of the expressions obtained for the statistical properties is investigated as a function of noise amplitude. For given parameters, analytical results are in good agreement with simulation until a certain critical noise amplitude is reached, which can be estimated. The role of trigger zones in increasing the reliability of transmission is discussed.

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133. Stimulus Representation and the Timing of Reward-Prediction Errors in Models of the Dopamine System

Elliot A. Ludvig, Richard S. Sutton, E. James Kehoe

https://doi.org/10.1162/neco.2008.11-07-654 | pp. 3034-3054

The phasic firing of dopamine neurons has been theorized to encode a reward-prediction error as formalized by the temporal-difference (TD) algorithm in reinforcement learning. Most TD models of dopamine have assumed a stimulus representation, known as the complete serial compound, in which each moment in a trial is distinctly represented. We introduce a more realistic temporal stimulus representation for the TD model. In our model, all external stimuli, including rewards, spawn a series of internal microstimuli, which grow weaker and more diffuse over time. These microstimuli are used by the TD learning algorithm to generate predictions of future reward. This new stimulus representation injects temporal generalization into the TD model and enhances correspondence between model and data in several experiments, including those when rewards are omitted or received early. This improved fit mostly derives from the absence of large negative errors in the new model, suggesting that dopamine alone can encode the full range of TD errors in these situations.

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134. An Ongoing Subthreshold Neuronal State Established Through Dynamic Coassembling of Cortical Cells

Osamu Hoshino

https://doi.org/10.1162/neco.2008.08-07-589 | pp. 3055-3086

Ensemble activation of neurons, triggered or spontaneous, sometimes involves a common (overlapping) neuronal population known as core cells. It is speculated that the core cells functioning as a core nucleus have a role in dictating noncore cells' behavior and thus overall local network dynamics. However, the truth and its significance in neuronal information processing still remain to be seen. To address this issue, a neural network model of an early sensory cortical area was simulated. In the network model, noncore cells that have selective responsiveness to sensory features constituted noncore cell assemblies. Core cells, having unselective responsiveness, constituted a single core cell assembly. Sensory stimulation activated neuronal ensembles that were indistinguishable from those activated spontaneously. The core cells were active in every ensemble activation and recruited a changing complement of noncore cells, which varied from spontaneous event to spontaneous event or from triggered event to triggered event. Ensemble activation of neurons was established through what we call dynamic coassembling, in which the core cell assembly and one of the noncore cell assemblies were dynamically linked together. Transient dynamic coassembling frequently and randomly took place during the ongoing (spontaneous) neuronal activity period, and persistent dynamic coassembling did during the stimulus-triggered neuronal activity period. The frequent ongoing activation of core cells mediated through transient dynamic coassembling depolarized

noncore cells just below firing threshold, whereby the noncore cells could respond rapidly to sensory stimulation. The persistent dynamic coassembling enhanced the responsiveness of noncore cells. We suggest that the core cells, functioning as a core nucleus, dictate how the noncore cells oscillate at a subthreshold level during the ongoing period and how to respond when stimulated. The transient and persistent dynamic coassembling may be an essential neuronal mechanism for the cortex to prepare and respond effectively to sensory input.

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135. Optimal Approximation of Signal Priors

Aapo Hyvärinen

https://doi.org/10.1162/neco.2008.10-06-384 | pp. 3087-3110

In signal restoration by Bayesian inference, one typically uses a parametric model of the prior distribution of the signal. Here, we consider how the parameters of a prior model should be estimated from observations of uncorrupted signals. A lot of recent work has implicitly assumed that maximum likelihood estimation is the optimal estimation method. Our results imply that this is not the case. We first obtain an objective function that approximates the error occurred in signal restoration due to an imperfect prior model. Next, we show that in an important special case (small gaussian noise), the error is the same as the score-matching objective function, which was previously proposed as an alternative for likelihood based on purely computational considerations. Our analysis thus shows that score matching combines computational simplicity with statistical optimality in signal restoration, providing a viable alternative to maximum likelihood methods. We also show how the method leads to a new intuitive and geometric interpretation of structure inherent in probability distributions.

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136. Sleeping Our Way to Weight Normalization and Stable Learning

Thomas J. Sullivan, Virginia R. de

https://doi.org/10.1162/neco.2008.04-07-502 | pp. 3111-3130

The functions of sleep have been an enduring mystery. Tononi and Cirelli (2003) hypothesized that one of the functions of slow-wave sleep is to scale down synapses in the cortex that have strengthened during awake learning. We create a computational model to test the functionality of this idea and examine some of its implications. We show that synaptic scaling during slow-wave sleep is capable of keeping Hebbian learning in check and that it enables stable development. We also show theoretically how it implements classical weight normalization, which has been in common use in neural models for decades. Finally, a significant computational limitation of this form of synaptic scaling is revealed through computer simulations.

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137. Volume 20 Index by Author

(authors unknown)

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No abstract available.

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138. Persistent Neural States: Stationary Localized Activity Patterns in Nonlinear Continuousn-Population,q-Dimensional Neural Networks

Olivier Faugeras, Romain Veltz, François Grimbert

https://doi.org/10.1162/neco.2008.12-07-660 | pp. 080804143617793-41

Neural continuum networks are an important aspect of the modeling of macroscopic parts of the cortex. Two classes of such networks are considered: voltage and activity based. In both cases, our networks contain an arbitrary number, n, of interacting neuron populations. Spatial nonsymmetric connectivity functions represent cortico-cortical, local connections, and external inputs represent nonlocal connections. Sigmoidal nonlinearities model the relationship between (average) membrane potential and activity. Departing from most of the previous work in this area, we do not assume the nonlinearity to be singular, that is, represented by the discontinuous Heaviside function. Another important difference from previous work is that we relax the assumption that the domain of definition where we study these networks is infinite, that is, equal to R or R2. We explicitly consider the biologically more relevant case of a bounded subset omega of Rq, q = 1, 2, 3, a better model of a piece of cortex. The time behavior of these networks is described by systems of integro-differential equations. Using methods of functional analysis, we study the existence and uniqueness of a stationary (i.e., time-independent) solution of these equations in the case of a stationary input. These solutions can be seen as 'persistent'; they are also sometimes called bumps. We show that under very mild assumptions on the connectivity functions and because we do not use the Heaviside function for the nonlinearities, such solutions always exist. We also give sufficient conditions on the connectivity functions for the solution to be absolutely stable, that is, independent of the initial state of the network. We then study the sensitivity of the solutions to variations of such parameters as the connectivity functions, the sigmoids, the external inputs, and, last but not least, the shape of the domain of existence omega of the neural continuum networks. These theoretical results are illustrated and corroborated by a large number of numerical experiments in most of the cases 2 < or = n < or = 3, 2 < or = q < or = 3.

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139. Spike-Timing Error Backpropagation in Theta Neuron Networks

Sam McKennoch, Thomas Voegtlin, Linda Bushnell

https://doi.org/10.1162/neco.2008.09-07-610 | pp. 080804143617793-37

The main contribution of this letter is the derivation of a steepest gradient descent learning rule for a multilayer network of theta neurons, a one-dimensional nonlinear neuron model. Central to our model is the assumption that the intrinsic neuron dynamics are sufficient to achieve consistent time coding, with no need to involve the precise shape of postsynaptic currents; this assumption departs from other related models such as SpikeProp and Tempotron learning. Our results clearly show that it is possible to perform complex computations by applying supervised learning techniques to the spike times and time response properties of nonlinear integrate and fire neurons. Networks trained with our multilayer training rule are shown to have similar generalization abilities for spike latency pattern classification as Tempotron learning. The rule is also able to train networks to perform complex regression tasks that neither SpikeProp or Tempotron learning appears to be capable of.

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140. Generation of Correlated Spike Trains

Romain Brette

https://doi.org/10.1162/neco.2008.12-07-657 | pp. 080804143617793-28

No abstract available.

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141. A Master Equation Formalism for Macroscopic Modeling of Asynchronous Irregular Activity States

Sami El■Boustani, Alain Destexhe

https://doi.org/10.1162/neco.2008.02-08-710 | pp. 080804143617793-55

International audience

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142. Long-Range Out-of-Sample Properties of Autoregressive Neural Networks

Patrick L. Leoni

https://doi.org/10.1162/neco.2008.09-07-615 | pp. 080804143617793-8

No abstract available.